

DYNAMICS OF TRAVERSABLE WORMHOLE BINARIES: MERGERS, GRAVITATIONAL WAVES, AND HEAVY BLACK HOLE SEEDS

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We present the first numerical-relativity evolutions of wormhole binaries: pairs of massive Ellis–Bronnikov drainholes held open by a ghost scalar field, evolved in 3+1 dimensions at equal masses, one throat compactness and zero spin. Each throat is an unstable fixed point whose e-fold time approaches the González–Guzmán–Sarbach linear rate with resolution, to 2.5% at our finest level, so its lifetime grows only as the logarithm of its initial perturbation. Two throats obey a charge-sign rule: like scalar charges repel and opposite charges attract, with a measured attraction-to-repulsion ratio of 1.462 ± 0.022 against the predicted $3/2$, so only opposite-signed pairs merge. They do not fuse into one wormhole. A head-on collision forms a common marginally outer trapped surface around both still-open throats, born with their combined area; its irreducible mass exceeds the ADM mass, so the Penrose inequality fails for this matter. The horizon then loses more than a quarter of its area as it swallows the phantom field and settles near the Schwarzschild radius of the pair’s mass: a black hole with two wormholes inside it. An orbital plunge ends in a remnant of the same size. With more than 50–70% of the circular momentum the pair passes without merging; the fly-by forms no horizon, and both mouths inflate. Gravitational waves are measured from both conversions. A lone collapsing throat sheds its seeded quadrupole as a burst ringing at 0.55 of the equal-mass Kerr frequency—following the first gravitational-wave signal reported from a wormhole collapse, the massless throat’s [44]—and every encounter radiates its own burst of gravitational waves and a scalar dipole of comparable energy and negative sign, which a horizon switches off on the timescale on which the remnant’s mass settles. A throat survives at most a few orbital periods for any plausible perturbation, and the negative-energy dipole would pump rather than drain the orbit, so these binaries have no inspiral. Their signal is a single burst whose frequency falls or drifts where a Kerr ringdown would hold; a search of 2.26 h of LIGO O3b data finds none. If such throats formed as late as $z \approx 20$, their collapse would supply halo-independent heavy black-hole seeds and millihertz bursts resolvable by LISA. The throats that inflate would create expanding anti-trapped regions from regular data, which the null energy condition otherwise forbids.

I. INTRODUCTION

Wheeler’s spacetime foam is a sea of handles—wormholes that open and close on the Planck scale [1–3]. Several mechanisms promote them to macroscopic size: inflation stretches a Planck-scale Lorentzian wormhole with the background [4]; bubble and domain-wall cosmologies produce post-inflationary regions joined through wormholes [5, 6, 92, 93]; and a gas of relic wormholes has been studied as a cosmology in its own right [7, 8]. The bubble and wall wormholes are not traversable and pinch off into black holes [6, 92]; none of this work evolves a traversable relic—what one does, and what two of them do to each other. That is the subject of this paper.

Two classical facts govern the dynamics. First, the simplest evolvable traversable wormhole—the Ellis–Bronnikov drainhole held open by a ghost scalar [9–12]—is unstable [15, 64, 88]: depending on the sign of the perturbation, the throat collapses to a black hole or inflates, on a timescale set by the throat radius [13, 14, 76, 87, 89, 91], as 3+1 evolutions of the

massless throat confirm nonlinearly [44]. A ghost field is a classical model here: as a quantum field coupled to gravity it destabilises the vacuum unless its effective theory is cut off at low energy with Lorentz violation [74], and we treat it as a classical phenomenological model throughout. Second, Geroch’s theorem forbids classical topology change [16, 17]: two wormholes cannot fuse into one, and none can be created classically. Creation is quantum [18], so our evolutions start at $t = 0$ from the newborn state—the stipulated initial data of every numerical binary, here with a physical origin attached.

Together these facts set selection rules. No astrophysical assembly channel exists: a throat decays faster than dynamical friction or radiation reaction could pair it. A primordial pair born from foam needs no assembly—birth and collision coincide, at a separation comparable to the objects—and its evolution is a race between orbital collision and throat decay, both rates fixed by the classical solution. The scalar adds a second rule: a phantom field repels like charges and attracts opposite ones, so like-signed pairs scatter while opposite-signed pairs fall together; and the sign of a lone throat’s perturbation selects collapse—a wormhole-to-primordial-black-hole channel beside the standard ones [19–21]—or inflation. A foam population thus splits into four fates—merger, scattering, collapse, inflation—across birth sepa-

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ration, perturbation and relative sign, the space mapped here. The binary evolved has three asymptotic ends: two independent wormholes, each leading to its own far universe. A foam handle differs—its two mouths belong to one wormhole and carry opposite scalar charge by construction—so our opposite-signed pairs reproduce a handle’s charges but not its topology. What merges is not two wormholes into one but two wormholes into one black hole containing both throats.

The observable is not a chirp but a population of bursts from the merger epoch, redshifted into whatever band the mass scale selects [22–25]. The epoch-independent deliverable is the energy radiated per event and its spectrum in throat-mass units (Sec. IX B). Every waveform behind it is extracted in the near zone, on spheres at $R = 10\text{--}44$, and none is extrapolated to null infinity: each radiated energy carries the spread across its spheres as a near-zone error bar (Sec. VIII). Collapse, alone or in a merger, also bears on JWST’s overmassive black holes at $z \gtrsim 4\text{--}6$, which favour heavy seeds of $10^4\text{--}10^6 M_\odot$ from formation channels that remain contested [46–49, 52]; Sec. X B explores what a population of such throats would imply for heavy seeds and millihertz bursts, conditional on the epoch at which they form, which our evolutions do not fix.

Sections II and III give the model and numerics. Sections IV–VII follow the campaign from one throat to two: the lone wormhole, two throats at rest, and the head-on and spiral mergers. Section VIII treats the radiation of every channel, and Sec. IX confronts the rescaled waveforms with LIGO–Virgo–KAGRA data. Section X draws the consequences and states the scope; Sec. XI gives reproducibility and cost. Appendix A collects the code-health record—constraint evolution, the interior failure under refinement and gauge, and device checks—and Appendix B supplementary measurements. All evolutions use GRTeclyn [41], the AMReX-based [42] GPU port of GRChombo [43], continuing Refs. [44, 45]. We use $G = c = 1$.

II. THE MODEL

A. Field equations

The matter is a single real, massless phantom scalar: gravity couples to *minus* the canonical stress tensor,

$$T_{ab} = -\left(\nabla_a \phi \nabla_b \phi - \frac{1}{2} g_{ab} \nabla_c \phi \nabla^c \phi\right), \quad V(\phi) \equiv 0, \quad (1)$$

so the energy density seen by normal observers, $\rho = -\frac{1}{2}(\Pi^2 + \gamma^{ij} \partial_i \phi \partial_j \phi)$ with $\Pi = (\partial_t \phi - \beta^i \partial_i \phi)/\alpha$, is non-positive everywhere, and the null energy condition fails wherever ϕ varies—the exotic stress that holds a throat open. The field itself obeys the canonical wave equation $\square \phi = 0$, evolved in first-order form in ϕ and Π .

B. Drainhole initial data

Each body is a massive Ellis–Bronnikov drainhole [9, 10]. In isotropic radius r about the throat centre, with

$$X = \frac{r - a^2/4r}{a}, \quad \Omega = 1 + \frac{a^2}{4r^2}, \quad u = \frac{m}{a} \left(\arctan X - \frac{\pi}{2} \right), \quad (2)$$

the exact static solution of Eq. (1) is

$$\alpha = e^u, \quad \gamma_{ij} = e^{-2u} \Omega^2 \delta_{ij}, \quad \phi = C \arctan X, \quad (3)$$

with $4\pi a^2 C^2 = a^2 + m^2$, $K_{ij} = 0$, $\Pi = 0$ and zero shift; in the evolved conformal variables $\chi = e^{2u} \Omega^{-2}$ and $h_{ij} = \delta_{ij}$. Both constraints hold identically at $t = 0$. Here a is the throat scale and m the ADM mass, carried by the lapse rather than the conformal factor, so χ at the throat stays in $[0.15, 0.25]$ for $m/a \leq 1$ and mass costs no resolution. The minimal surface sits at $r_t = \frac{1}{2}(m + \sqrt{m^2 + a^2})$, where $X = m/a$, so its areal radius is analytic: $R_{\min} = e^{-u(r_t)} \sqrt{a^2 + m^2} = \sqrt{a^2 + m^2} \exp[(m/a) \arctan(a/m)]$, which is a at $m = 0$. The production throat $a = 2$, $m = 1$ has $R_{\min} = \sqrt{5} e^{\frac{1}{2} \arctan 2} = 3.8895$ at $r_t = \frac{1}{2}(1 + \sqrt{5}) = 1.618$. The point $r \rightarrow 0$ is the far side’s spatial infinity, compactified ($\chi \sim r^4$). At $m = 0$ the throat is ultrastatic ($\alpha \equiv 1$, $M_{\text{ADM}} = 0$) and two such throats do not attract.

C. The binary superposition

Two throats at separation d (one-body radii r_A , r_B) are superposed in excess form,

$$u = u_A + u_B, \quad \psi = 1 + (\sqrt{\Omega_A} - 1) + (\sqrt{\Omega_B} - 1), \quad (4)$$

with $\chi = e^{2u} \psi^{-4}$, $\alpha = e^u$ and $\phi = \phi_A + \sigma \phi_B$, $\sigma = \pm 1$; the constant tail of the arctan profiles is subtracted so that $\phi \rightarrow 0$ at the boundary. In the superposed campaign momentum entered through one Bowen–York term [35] per throat,

$$\hat{A}_{ij} = \frac{3}{2r^2} [P_i n_j + P_j n_i - (\delta_{ij} - n_i n_j) P \cdot n], \quad (5)$$

converted by $\tilde{A}_{ij} = \chi^{3/2} \hat{A}_{ij}$, with $K = 0$; the production campaign instead carries momentum with the exact boost of Sec. II E. In either case the throats sit at $x = \mp d/2$ with $P^i = (0, \pm p, 0)$: momenta of magnitude p in units of the one-body mass, tangential to the separation and opposite on the two throats, so the pair carries no net momentum and orbital angular momentum $J = pd$; $p = 0$ is a release from rest. Since $\Pi = 0$ the matter momentum density vanishes and $\hat{A}_{ij}$ is flat-space divergence free, so the momentum constraint holds *exactly* at $t = 0$; only the Hamiltonian constraint carries the superposition defect, $O(a^2/d^2) + O(m/d)$. The relative sign σ is the pair’s one discrete parameter. Far from its throat each body is a mass m carrying the scalar charge $q = Ca$ ($\phi_A \simeq -q/r_A$ once the constant tail is

subtracted), and a phantom mediator *repels* like charges and *attracts* opposite ones, so two throats at rest a large distance d apart attract with

$$F = \frac{m^2}{d^2} (1 - \sigma Q), \quad Q \equiv \left| \frac{F_\phi}{F_{\text{grav}}} \right| = \frac{4\pi q^2}{m^2} = \frac{a^2 + m^2}{m^2} > 1 \quad (6)$$

always ($Q = 5$ for the production throat): a like-signed pair repels and never merges under its own gravity, and $\sigma = -1$ is the merger channel. A windowed one-body correction of the Helfer type [36] was tested and not adopted: it lowers the $t = 0$ defect but moves it into the fade zone between the throats, where it stalls the orbit instead of letting it plunge. The superposition itself is not evolved for physics: it seeds the constraint solve of Sec. II D, which removes the defect and fixes which throats the pair is made of, and the superposed campaign that preceded the solve is kept as the systematics study (Table III).

D. Constraint-solved data and matched placement

With $\Pi = 0$, $K = 0$ and $\gamma_{ij} = \Psi^4 \delta_{ij}$ the Hamiltonian constraint of Eq. (1) reads, in flat operators,

$$\nabla^2 \Psi - \pi |\nabla \phi|^2 \Psi + \frac{1}{8} \hat{A}_{ij} \hat{A}^{ij} \Psi^{-7} = 0. \quad (7)$$

The single drainhole solves it exactly; the superposition does not. We solve it for a correction w to the superposed Ψ on the whole initial mesh hierarchy with AMReX's multigrid [42], holding ϕ , $\hat{A}_{ij}$, Π and the lapse fixed so that the momentum constraint stays exact (a Newton iteration on the Bowen–York term; a Robin outer boundary). On the $d = 8$ head-on the Hamiltonian on the finest level's throat shell falls from 9.7×10^{-3} to 5.1×10^{-6} rms, the exact throat's discretisation floor. Evolved from solved data at the superposition's c , the $d = 8$ head-on merges as the superposed one does, its common horizon forming about two units later and 4–5% larger; its $M_{\text{ADM}} = 2.738$ puts $R/2$ of that horizon 7–8% above the pair's mass.

Which throat is each mouth? Near its centre $\Psi = c/r + d_c + O(r)$, and the inversion $r' = c^2/r$ turns this into an asymptotically flat end—the mouth's own far universe—with ADM mass and scalar charge

$$M_{\text{far}} = 2c d_c, \quad Q_{\text{far}} = \frac{4C c^2}{a}, \quad (8)$$

both frozen in time, like a puncture's individual mass. The isolated drainhole has $M_{\text{far}} = -m e^{\pi m/a} = -4.8105$ and $Q_{\text{far}} = \sqrt{a^2 + m^2} e^{\pi m/a} / \sqrt{4\pi} = 3.0344$. A pair $(M_{\text{far}}, Q_{\text{far}})$ names one isolated drainhole, and its near-side mass m' is the mouth's one-body mass.

The superposition does not place isolated throats (Table I). Each throat inherits the companion's constants, $\Psi \rightarrow e^{-u_B(x_A)/2} \Psi$ near A , and is rescaled: at $d = 8$ each mouth has one-body mass $m' = 1.149$ and $R_{\text{min}} = 4.466$, 14.8% above $R_\star$ (the placement excess of Sec. V C),

TABLE I. Each mouth of the $d = 8$ pair at $t = 0$ ($L = 64$, level 3): minimal areal radius, far-side mass [Eq. (8)], one-body mass m' , and the pair's M_{ADM} from Eq. (9). The superposed slice is not a solution; its 2.00 is read from its tail.

data	R_{min}	M_{far}	m'	M_{ADM}
isolated throat	3.890	−4.810	1	1.001
superposed	4.466	−5.363	1.149	2.00
solved, superposed c	4.523	−5.368	1.148	2.738
solved, matched	3.892	−4.810	1	2.362

and at $d = 12$ it has $m' = 1.094$. Solving Eq. (7) at the superposition's puncture coefficients keeps these mouths— M_{far} moves by 0.1% and R_{min} by 1.3%—so a solved run and its superposed twin differ only in the data. Because ϕ fixes a static throat's coordinate size, c alone cannot remove the rescaling. The *matched placement* gives each throat a coordinate scale λ ($a \rightarrow \lambda a$ and $m \rightarrow \lambda m$ in ϕ , u and Ω ; C is scale-free), sets $\lambda = (c/c_{\text{iso}})^2$ with $c_{\text{iso}} = \frac{a}{2} e^{\pi m/2a}$, which holds Q_{far} at its isolated value, and iterates c until M_{far} is isolated too. Each mouth is then locally the isolated static drainhole: its R_{min} lands within 0.06% of $R_\star$ without being imposed, and $m' = 1$, so that p is in units of the true one-body mass. With momentum the iteration converges the same way ($d = 12$, $p = 0.12$: R_{min} within 0.02% of $R_\star$).

The pair's mass. With $\Psi \rightarrow 1 + M_{\text{ADM}}/2r$, Gauss's theorem on Eq. (7), whose punctures contribute $-4\pi c_X \delta_X$, gives

$$M_{\text{ADM}} = 2 \sum_X c_X - \frac{1}{2\pi} \int \left(\pi |\nabla \phi|^2 \Psi - \frac{1}{8} \hat{A}_{ij} \hat{A}^{ij} \Psi^{-7} \right) d^3 x, \quad (9)$$

which returns the exact throat's $M = 1$ to 0.14% and, unlike a fit to the monopole tail, does not read the outer boundary's offset of w as mass. For matched pairs at rest (Table II) the excess over two isolated throats follows

$$E_b \equiv M_{\text{ADM}} - 2m = -\lambda^2 \frac{m^2}{d} (1 + \sigma Q) \quad (10)$$

for both signs, to 3% at $d = 8$ and 0.5% at $d = 48$. It is the Newtonian binding $-\lambda^2 m^2/d$ plus the phantom cross energy $-\sigma \int \nabla \phi_A \cdot \nabla \phi_B d^3 x = -\sigma \lambda^2 (a^2 + m^2)/d$ of two charges λq , with $\lambda \rightarrow 1$ as $d \rightarrow \infty$; the solve adds nothing else. In the merger channel $\sigma = -1$ the scalar term is positive and wins: the attracting pair weighs *more* than its parts, $E_b d \rightarrow +4$, and the repelling pair less, $E_b d \rightarrow -6$. Read at fixed charges, Eq. (10) would make the $\sigma = -1$ pair repel. Equation (6) and the measured pull (Sec. V A) follow instead if the scalar part acts at fixed potential, $F_\phi = +\partial E_\phi / \partial d$, as between conductors held at fixed voltage: the value of ϕ at each throat's far infinity cannot change, while the flux through the throat can. The evolution has now answered: rest pairs rerun on matched data pull and push in the ratio 1.462 ± 0.022 (Sec. V A), the fixed-potential prediction, while fixed charges would reverse the sign rule itself.

TABLE II. Mass of matched pairs at rest [Eq. (9)] against two isolated throats on the same grid ($L = 128$, $\Delta x = 0.0625$ at the throats), and Eq. (10). λ : each throat’s coordinate scale.

d	$\sigma = -1$ (attract)			$\sigma = +1$ (repel)		
	λ	E_b	Eq. (10)	λ	E_b	Eq. (10)
8	0.857	+0.357	+0.367	0.906	−0.619	−0.616
12	0.906	+0.269	+0.274	0.931	−0.437	−0.433
16	0.931	+0.214	+0.217	0.946	−0.338	−0.335
24	0.955	+0.151	+0.152	0.962	−0.233	−0.231
48	0.979	+0.080	+0.080	0.980	−0.121	−0.120

Units of this paper. Masses and times are in units of the drainhole parameter $m = 1$. Every production binary evolves matched, solved data: each mouth’s far side is the isolated throat’s, $m' = 1$ to 6×10^{-6} across the production set, so a quoted momentum p is per true one-body mass, with boost speed $v = p/\sqrt{1+p^2}$, and the pair’s mass is the solve’s M_{ADM} [Eq. (9); 2.3573 for the $d = 8$ head-on pair]. The superposed campaign’s mouths were instead rescaled by the placement—they have $m' = 1.149$ at $d = 8$ and $m' = 1.094$ at $d = 12$, so the p it quotes runs from $p/m' = 0.110$ to 0.411 per one-body mass over $p = 0.12$ – 0.45 . The trajectories themselves are robust to the placement: rerun from matched data, the rest pairs keep the sign rule, its ratio and the force ladder (Secs. V A and V B).

E. Throats with momentum

A throat with ADM momentum p is the exact drainhole boosted to $v = p/\sqrt{m^2 + p^2}$ and cut at $t = 0$: the metric, K_{ij} , ϕ and Π together, with the boosted lapse and shift, so that the slice is an exact solution of both constraints and the scalar moves with its throat ($\Pi \neq 0$). On the slice every χ contour, the throat’s included, is an ellipsoid contracted along the motion by $1/\gamma = m/\sqrt{m^2 + p^2}$, as a boosted black hole’s horizon is; the measured axis ratios follow this law to 6×10^{-5} at every p up to 0.45 (Fig. 1). The alternative—Bowen–York momentum on round, conformally flat data with the scalar at rest, the black-hole recipe—is not a moving wormhole: at $p = 0.45$ it inflates the throat by 10% in R_{min} by $t = 20.2$, where the boosted throat holds its shape-fitted radius to 0.05% through $t = 16$.

A pair is built from two boosted throats as in Sec. II D, with the companion’s K_{ij} and Π windowed out inside each throat, and the solve then corrects both constraints, for w and a vector potential W ($\hat{A} \rightarrow \hat{A} + \text{LW}$). Three checks anchor it: on a lone boosted throat the solve is idle ($\max |w| \leq 5 \times 10^{-7}$, $\max |W| \leq 5 \times 10^{-6}$, the far side isolated to 4×10^{-7}); on the rest pair it reproduces the scalar solve’s c , σ and w_0 to 10^{-5} ; and on the flipped $d = 12$, $p = \pm 0.25$ pair on the production grid the far sides come out the isolated throat’s to 10^{-5} , each mouth’s

R_{min} lands 0.09% from the isolated value, and the throat-shell Hamiltonian falls from 7.1×10^{-3} to at most 7.8×10^{-5} rms across the refinement levels.

F. Declared seeds

The instability is probed with declared perturbations of the conformal factor only: a Gaussian shell on the minimal surface,

$$\psi \rightarrow \psi[1 + \varepsilon g(r)], \quad g = e^{-(r-r_t)^2/w^2}, \quad w = a/4, \quad (11)$$

which changes the throat’s areal radius by a fraction $\approx 2\varepsilon$, and a quadrupolar factor

$$\psi \rightarrow \psi[1 + \varepsilon_2 g(r) P_2(\cos \theta)] \quad (12)$$

about the z axis through the throat. Neither seed is constraint-solved. K_{ij} and Π stay untouched, so the momentum constraint holds at $t = 0$; but the drainhole solves the linear equation $\nabla^2 \Psi = \pi |\nabla \phi|^2 \Psi$ ($\Psi^4 = \chi^{-1}$, flat operators), so the spherical seed leaves exactly $\mathcal{H} = -8\varepsilon \Psi^{-4}(1 + \varepsilon g)^{-5}(\nabla^2 g + 2\nabla \ln \Psi \cdot \nabla g)$ —linear in ε , confined to the shell and, the shell being narrow, not small there: its peak is $0.93 \times 16\pi|\rho|$ at $|\varepsilon| = 10^{-2}$. The quadrupolar seed’s defect is pure $\ell = 2$ at first order. CCZ4 feeds the defect into Θ and, within the first steps, into $\mathcal{M}$ (Fig. 10(b)). Above $\varepsilon \simeq 0.025$ the spherical seed turns the minimal surface into a maximum between two minima. The spherical seed selects the branch but cannot make waves: a spherical collapse has no $\ell \geq 2$ moment and radiates no gravitational waves. The quadrupolar seed is there for that reason. It has no $\ell = 0$ part at first order, so it does not compete with the spherical seed for the branch, and it gives a collapsing throat a quadrupole to shed—the only gravitational-wave signal a lone throat has (Sec. VIII A).

III. NUMERICAL SETUP

The campaign is one-knob: every run differs from a named reference in a single setting, or repeats one with diagnostics, checkpoints or time added, and has one line—difference, outcome, caveats—in the registry shipped with the results (Sec. XI). Table III groups the 216 evolutions; 12 form an archived shakedown that fixed the gauge, dissipation and tagging choices below and are not used for physics, leaving 204.

A. Evolution scheme

We evolve CCZ4 [37] with damping $(\kappa_1, \kappa_2, \kappa_3) = (3, 0, 1)$, 1+log slicing [38] started from the exact static lapse of Eq. (3), a Gamma-driver shift [39, 40, 78] with $\eta = 1$ started from zero, fourth-order stencils, RK4 and

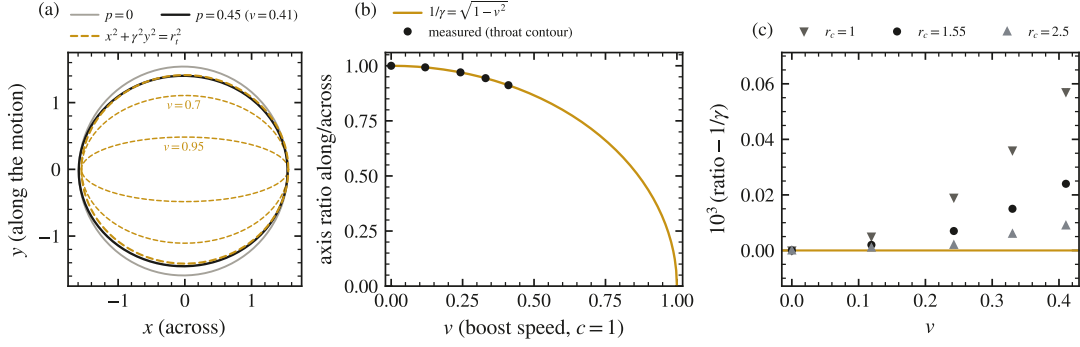


FIG. 1. The moving throat at $t = 0$ is the Lorentz-contracted drainhole. (a) χ contours of the throat at rest and at $p = 0.45$, the latter on its analytic ellipse $x^2 + \gamma^2 y^2 = r_t^2$; the fainter ellipses continue the law to $v = 0.7$ and 0.95 . (b) Axis ratio (along/across the motion) against the boost speed $v = p/\sqrt{m^2 + p^2}$, on $1/\gamma = \sqrt{1 - v^2}$ drawn to the disc limit $v = 1$. (c) The residual: below 6×10^{-5} at every v and at three contour levels. The outcome: the boosted data carry exactly the special-relativistic shape, with the scalar in motion, and both constraints hold at the exact solution's discretisation floor.

TABLE III. The run matrix: the 216 packed evolutions in groups, each varying one knob against a named reference or repeating one; per-run lines are in the shipped registry. Left: the production campaign on constraint-solved (mode-3) initial data, with momentum carried by the exact Lorentz boost (Sec. II E). Right: the superposed-data campaign it supersedes, kept as the systematics study; p there is each throat's tangential Bowen–York momentum (Sec. II C). Unless stated, $a = 2$, $m = 1$ ($R_{\min} = 3.89$), $\sigma = -1$, $d = 12$ and $L = 64$.

group	knob(s)	runs	Sec.	group	knob(s)	runs	Sec.
lone throat	levels 2–4; χ floor; Δt ; $L=128$	9	IV	<i>superposed-data campaign (archived): the systematics study, Sec. IID</i>			
spherical kicks	$\varepsilon = \pm 10^{-3} \dots \pm 10^{-1}$; levels 3–5	10	IV C	two throats at rest	σ ; $a = 1-3$; $d = 12-18$; $p = 0.12$ push	9	V
quadrupolar kicks	$\varepsilon_2 = 0.005-0.05$; kick-free arm	6	VIII A	placement probes	$d = 6-48$, one step each	18	V C
scalar-stream re-runs	unkicked level 4, $L=64/128$; $\varepsilon=+10^{-2}$	3	VIII G	head-on, $d = 8/12$	fill; levels 3/5; down-step; $\pm \varepsilon$	10	VI
boosted throat	$p = 0-0.45$ at $t=0$; collapse twins, levels 3/4; Bowen–York contrast	10	II E	half-mass, $m = 0.5$	pair at $d = 6$; lone control	2	VIC
constraint-solved data	GRtresa bridge; in-code solve: head-on scout; kicked single	3	IID	momentum scan	$p = 0.20-0.45$	4	VII A
solve verifications	boosted pairs at $t=0$ on the production grid; the pair at level 3 to $t=20$	3	II E	fly-by, $p = 0.45$	level 3; level 5 on the doubled box	2	VIII D
matched rest pairs	σ ; $a = 1-3$; $d = 12-18$	8	V	orbital chain, $p = 0.12$	N ; levels; damping; dissipation; χ floor	13	VII
matched placement	$d = 6-48$, one step each	18	V C	refinement ladder	levels 4–7; Δt ; damping	8	VII D
head-on, mode 3	legs, levels 4–6; MOTS replay; first-law window	8	VI	interior freeze, $L=64$	skin radii; arming time; levels; extraction	7	VIII C
merger chain, $d=6$	$p=0.10$ tangential; levels 3–7; σ_{KO} ; χ floor; settle leg	7	VII C	failure probes	$p = 0.15-0.25$; levels; fill off	6	VII D
$d=12$ arms, mode 3	$p = 0.12-0.25$: tangential, inward, Bowen–York	3	VII	production spiral, $L=128$	three-stage chain; fill-window twin; level 5 from $t=0$; mouth scans	6	VII C
fly-by and E_{GW} scan	$p = 0.25-0.90$; χ -floor and level-5 extensions	8	VIII D	Helper twins	correction; window; levels	6	II C
convergence, mode 3	spiral at level 4; wave-zone ball; dissipation	3	VIII	gauge arms	1+ log halved; harmonic-class; $\eta=4$; spiral, head-on	10	VII D
vacuum BBH	same d , p , masses; no scalar, no sponge	4	VIII E	shakedown (archived)	gauge, dissipation, tagging; boosted throat	12	III A

Kreiss–Oliger dissipation $\sigma = 0.1$. The time step is $\Delta t = 0.02 \Delta x$, an order of magnitude below the usual CCZ4 Courant factor, because larger steps fail: a multiplier of 0.05 departs from the production run at $t \approx 33$,

and 0.1 dies at $t = 16$. A halved step was run only on the spiral's interior failure (Sec. VII D); elsewhere time-discretisation error is not separately bounded, and every other resolution statement is at fixed $\Delta t/\Delta x$.

B. Grid, boundaries and regularisation

The domain is a cube of side $L = 64$ centred on the system, with a 128^3 base grid ($\Delta x = 0.5$). Nested boxes, halving per level and centred on the tracked throats (half-width 4 per throat at the production finest level), are redrawn every 16 steps. Production runs use three refinement levels (finest $\Delta x = 1/16$); resolution arms go to levels 4–7 where stated, and the production spiral, the level-5 fly-by and two lone-throat arms use $L = 128$: the same cells in a box twice as wide. Level ℓ thus means a finest cell $\Delta x = 0.5/2^\ell$. The one exception is the octant of an $L = 512$ box that carries the inflating throat (Sec. IV F): its base spacing is 2, so its 5 levels give $\Delta x = 1/16$, the resolution of our level-3 arms, and we quote it by that Δx . Outer boundaries are Sommerfeld, backed by a spherical sponge about the box centre over the outer quarter of the half-width— $r = 24$ –32 at $L = 64$ and 48–64 at $L = 128$ —in which extra dissipation ramps to $\sigma_{\text{sp}} = 4$ as the fourth power of the depth; every extraction sphere lies inside its inner radius. The sponge is Kreiss–Oliger dissipation only: it damps grid-scale noise but is transparent to long-wavelength gauge and physical waves, which reflect from the box faces (Sec. VI B). The vacuum controls (Sec. VIII E) instead run GRTeclyn’s binary-black-hole example with its own numerical settings and no sponge.

Each throat carries a compactified second infinity at its centre, where $\chi \sim r^4$, so the evolved variables are floored at $\chi \geq 10^{-8}$ and $\alpha \geq 10^{-10}$ (a variant floors χ at 10^{-20} and applies 10^{-8} only where $1/\chi$ is evaluated). Two further devices act only inside regions already trapped: exponential damping of the scalar where the lapse has collapsed, and an interior freeze that multiplies every right-hand side by $1 - W(r)$, with W a C^2 quintic step between radii r_{full} and r_{start} . All three are numerical devices, each measured by a twin: χ -floor twins at 10^{-8} and 10^{-20} re-converge to four digits after touching the floor, the scalar damping moves the spiral’s interior failure by less than a unit (Sec. VII D), and the freeze leaves the exterior unchanged (Secs. VI B and VII C).

C. Diagnostics

Every coarse step the code writes the volume-averaged norms $\|\mathcal{H}\| = (\int \mathcal{H}^2 dV/L^3)^{1/2}$ and $\|\mathcal{M}\| = (\int \mathcal{M}_i \mathcal{M}_i dV/L^3)^{1/2}$, evaluated with fourth-order differences on the base grid over the whole domain—sponge and throat interiors included—after the finer levels have been restricted onto it; they depend on the box and are compared only within one. These base-grid norms do not resolve the throats; they monitor code health, not local accuracy. Both throat centres are tracked, and about each the areal radius $R(r)$ is scanned on coordinate spheres; its minimum is the throat. The spheres run out to $r = 2.79$, a fixed setting of the offline instrument: wide enough for the static throat and the collapse horizons, but an inflating throat can pass it, after which

the edge reading bounds the throat’s areal radius from above. On the same spheres the null expansions $\theta_\pm$ are computed with the outward direction fixed by *increasing areal radius*, so that a dynamic throat is marginally *anti-trapped* [33, 76]. The orientation matters: the in-code monitor fixes the outward direction by increasing *coordinate* radius, which points the wrong way beyond a throat and manufactures trapped verdicts wherever $R(r)$ falls. Every horizon statement here is therefore read off-line from archived plotfiles with the areal orientation; the in-code columns serve only as a live monitor.

The star-shaped scan reports the outermost coordinate sphere that is trapped everywhere ($\max \theta_+ = 0$, $\theta_- < 0$). A trapped sphere lies inside the apparent horizon, so this sphere—“the scan’s MOTS” below—bounds the marginally outer trapped surface from inside. We quote its areal radius R and its Misner–Sharp mass [32],

$$\frac{2M_{\text{MS}}}{R} = 1 + \frac{R^2 \langle \theta_+ \theta_- \rangle}{4}, \quad (13)$$

with the product averaged over the sphere and $\theta_\pm$ computed for the null normals $k^a = n^a + s^a$ and $l^a = n^a - s^a$ (n^a the slice’s unit normal, s^a the sphere’s outward unit normal, so $k \cdot l = -2$); with the normalisation $k \cdot l = -1$ the factor $1/4$ becomes $1/2$. On an exact MOTS $M_{\text{MS}} = R/2$; on the scan’s spheres M_{MS} exceeds $R/2$ by at most 3% about a lone collapsed throat, but about a merged remnant by up to 34% while the surface forms (44% at level 3), falling to 4% as it rounds. This is the shape systematic of star-shaped scanning, and it also affects whether a surface is found at all. Because the phantom violates the null energy condition, a trapped surface need not lie inside an event horizon; “horizon”, “censored” and “black hole” below refer to MOTSs on the evolved slices. A deformable instrument backs the scan: a spectral flow finder expands the trial surface as $r = h(\theta, \phi)$ in real harmonics up to degree $\ell_{\text{max}} = 6$ (8 where stated) and drives its expansion to zero by a damped fast flow [77], with the same orientation rule. It recovers Kerr–Schild Schwarzschild ($R = 2M$ to 0.24% and $M_{\text{MS}} = M$ to 0.12% from spherical and dented seeds), finds no trapped surface outside the time-symmetric Ellis throat (whose minimal surface has $\theta_+ = \theta_- = 0$), and reproduces the star-shaped scan’s surface on the head-on remnant.

Radiation is read from Ψ_4 on spheres at $R = 10, 14, 18$ (and 22 on the single-throat wave arms) on the head-on and the single throats, whose $L = 128$ arms keep these spheres, and at $R = 20, 28, 36, 44$ on the $L = 128$ spiral and fly-by, decomposed in spin-weight -2 harmonics. Every amplitude quoted is $r\Psi_4$; on the head-on the in-code and offline pipelines agree at the burst peak to 0.2%, while on the single throats the in-code stream is dominated by its grid-scale floor and only the offline one is used. A scalar stream is read on the same spheres, and at $R = 14$ and 30 on the $L = 128$ spiral and fly-by: ϕ and Π in spin-weight-0 harmonics, $A_{\ell m} = R\phi_{\ell m}$, and the kinematic flux $F_{\text{kin}} = -R^2 \oint \Pi \partial_r \phi d\Omega$, positive for an outgoing *canonical* wave (Sec. VIII F). A per-cell

autopsy records the location and one-step history of any floating-point failure.

D. Marginally outer trapped surfaces

A marginally outer trapped surface (MOTS) is a closed two-surface of the slice on which the outgoing null expansion vanishes, $\theta_+ = 0$, while the ingoing one is negative, $\theta_- < 0$; the outermost MOTS is the apparent horizon. Unlike the event horizon, which is defined only once the entire future of the spacetime is known, a MOTS is quasi-local—it is read off a single slice—which makes it the operational horizon of numerical relativity. Because the phantom violates the null energy condition, a MOTS here need not lie inside an event horizon (Sec. III C); every “horizon”, “censored” and “black hole” in this paper is a statement about MOTSs on the evolved slices.

The evolution code does not locate horizons; finding a MOTS is an elliptic problem on a single slice and is solved in post-processing, on every archived slice, so horizon coverage matches the plot cadence. The finder expands the surface about the tracked centre as $r = h(\theta, \phi)$ in real spherical harmonics up to degree $\ell_{\max} = 6$ on the head-on slices and $\ell_{\max} = 8$ on the orbital ones, and solves $\theta_+[h] = 0$ by Newton iteration on metric and extrinsic-curvature data interpolated from the finest refinement level covering the surface, with the areal orientation of Sec. III C. The first surface of a run is found by the damped fast flow [77]; each later slice warm-starts from the previous surface and converges in a few iterations to a root-mean-square residual below the solver tolerance of 10^{-6} . From the converged surface we quote the areal radius $R = \sqrt{A/4\pi}$, the Misner–Sharp mass of Eq. (13), and the harmonic content beyond $\ell = 0$ as the deformation. The instrument recovers Kerr–Schild Schwarzschild ($R = 2M$ to 0.24%, $M_{\text{MS}} = M$ to 0.12%) and finds no trapped surface outside the time-symmetric Ellis throat. Every horizon number in this paper—formation times, R , M_{MS} —is read from these surfaces. The star-shaped scan of Sec. III C is retained only as an inner bound and a live monitor: a fully trapped round sphere lies strictly inside a deformed MOTS, so while the surface is forming the scan under-reads it by the shape systematic quoted there, and can miss it altogether.

IV. THE SINGLE WORMHOLE

A. The static throat as a fixed point

Evolved without a declared perturbation, the exact drainhole holds $R_\star = 3.8895$ to 0.1% for 35 time units at level 3 and to 1% until $t \simeq 44$ (Fig. 2(a)); at level 4 it is exact to four digits at $t = 30$. The departure moves *later* by 5.6–8.7 units per halving of the cell—a truncation-noise seed of fixed growth rate handed a smaller amplitude. Level 2 dies of an origin failure at $t = 24.2$ with the

throat still exact to 0.12%. A lighter throat is less stable: at level 3 the $m = 0.5$ throat holds 0.1% only to $t \approx 17$, against 35 at $m = 1$, and dies at its origin at $t = 26.3$.

B. The unstable mode

The departure is a single exponential mode (Fig. 2(a,b)): plateau fits give $\tau = 5.88$ (level 3) and 5.26 (level 4), i.e. $T \equiv \tau \alpha_{\text{th}}/R_\star = 0.87$ and 0.78, where $\alpha_{\text{th}} = e^{u(r_\star)} = 0.575$ is the static lapse at the throat, so that T is the proper-time e-fold in units of the throat’s light-crossing time. Our drainhole is the $\gamma_1 = m/a = 0.5$ member of the static family of González, Guzmán and Sarbach (their metric with $b = a$, $\gamma_0 = -\pi\gamma_1/2$), for which their linear analysis gives $T = 0.758$; their Table I spans $T = 0.846$ at the massless end to 0.590 [15]. Solving the linearised field equations of this solution ourselves—their regularised master equation and, independently, the reduction in the gauge $\delta\phi = 0$ —gives $T = 0.7584$, i.e. $\tau_{\text{lin}} = 5.13M$. The measured e-fold approaches it with resolution: it is 2.5% longer at level 4 and 15% longer at level 3; with two resolutions we quote no convergence order. The seed is truncation error, and its sign flips with resolution—level 3 collapses, level 4 inflates—confirming from exact massive data the two branches found for the massless throat [44].

C. Branching: collapse versus inflation

A declared kick, Eq. (11), overrides the truncation seed (Fig. 2(c)). At $\varepsilon = \pm 10^{-2}$ and $\pm 10^{-3}$ the fate is always opposite to the push, each $\pm$ pair crosses at the ε -independent $t_\times \simeq 13.0$, and a tenfold smaller kick arrives 11–12 units later at matched horizon radius—linear response on the same mode. The branching survives refinement: at level 4, whose own seed inflates the unkicked throat, $+10^{-2}$ still forms a horizon at $t = 11$ and -10^{-2} still inflates, the twins crossing at $t = 12.99$ against level 3’s 13.01. The branch point is the throat’s, not the grid’s or the kick’s. The kick also sets the clock: horizons form at $t = 11$ ($\varepsilon = +10^{-2}$) and $t = 25$ ($+10^{-3}$), against $t \leq 61$ for the unkicked throat (its first scanned slice after departure already holds a MOTS). At $\varepsilon = \pm 0.1$ the data are no longer perturbative (Fig. 2(c)): $+0.1$ turns the minimal surface into a maximum between two minima and is trapped from the first scan ($t = 1$), while -0.1 , which narrows the throat to $R_{\min} = 3.15$, re-expands to 3.36 by $t = 4$ and then collapses as well, trapped from $t = 6$ —against the sign rule. Both die at the compactified origin ($t = 14.1$ and 15.2 , NaN in h_{11}), the failure that also ends the $+10^{-3}$ arm at $t = 40.1$, before the lapse has collapsed anywhere. Their Hamiltonian defect is $3\text{--}6\times$ that of the 1% arms in the box norm and $8.6\text{--}10.3 \times 16\pi|\rho|$ on the shell; the usable range ends at 10^{-2} .

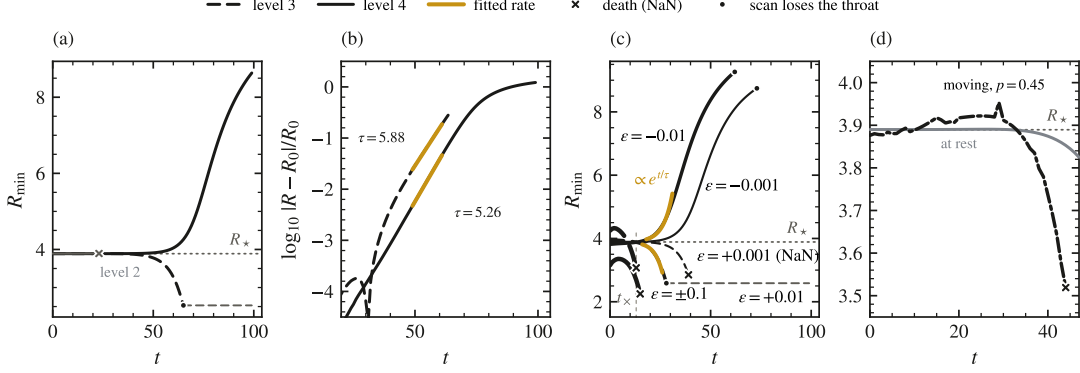


FIG. 2. Instability of the isolated throat. (a) Minimal areal radius from exact static data at refinement levels 2–4. The exact solution (R_* , dotted) is held until truncation noise excites the unstable mode: level 3 (dashed) collapses, level 4 (solid) inflates, level 2 (grey) dies of an origin failure first (cross). Dots mark where the scan loses the throat; muted lines carry the last reading. (b) Deviation from R_* : one exponential mode with a level-dependent sign, $\tau = 5.88/5.26$ at levels 3/4. (c) Level 3 with declared seeds, Eq. (11): $\epsilon = \pm 10^{-2}$ (heavy), $\pm 10^{-3}$ (light) and, heaviest, ± 0.1 , which both collapse; dashed, outward kicks. Twins cross at $t_{\times} \simeq 13.0$ independent of ϵ . Gold: exponential fits; crosses: deaths at the compactified origin, $t = 40.1$ ($+10^{-3}$), 14.1 ($+0.1$) and 15.2 (-0.1). Their constraint norms are Fig. 10(a,b). Unless a seed is declared, the grid’s truncation noise picks the branch. (d) The throat in motion (Sec. IV D): the exact Lorentz-boosted throat (dash-dot, $p = 0.45$, level 3) against the resting level-3 arm (grey). It rides R_* to $t \approx 29$ and then collapses on the level-3 mode, dying at $t = 44.67$: the fate and its rate are the resting throat’s, carried with the motion.

D. The throat in motion

A uniformly moving throat keeps the resting throat’s fate. The exact-boost $p = 0.45$ throat (Sec. II E; level 3, $v = 0.41$) rides R_* , rising at most 2.0%, until $t = 36.8$, where it falls 1% below its start (the resting level-3 throat: $t \simeq 44$), and then collapses on the resting throat’s own mode: $\tau = 5.5$ – 5.7 against 5.88 at rest, on the same branch—level 3 collapses—and it dies at the same numerical endpoint ($t = 44.67$, Fig. 2(a)). The instability is the throat’s, carried with the motion; the momentum changes neither the branch nor the rate.

E. The collapse endpoint

The collapse branch ends in a black hole that consumes its supporting field. On the unkicked level-3 arm the horizon’s areal radius falls $3.23 \rightarrow 2.37$ and $M_{\text{MS}} 1.62 \rightarrow 1.23$ over 39 units, with the origin lapse at 0.016; the kicked $\epsilon = +10^{-2}$ arm runs clean to $t = 100$ behind its horizon. Unlike the massless throat of Ref. [44], which bounced at $t \approx 4M$ and shed its horizon by $18.5M$, no arm here sheds a horizon. Ref. [44] drove that collapse by rescaling the phantom’s source term, halving its stress-energy in the Einstein equations; where the present unmodified evolutions differ—the bounce and the horizon shedding—they supersede it. The pure-quadrupole arm ($\epsilon_2 = 10^{-2}$, no radial kick) is fully instrumented (Fig. 3): a MOTS from $t = 33$ at $R = 3.80$, shrinking to 2.41 ($M_{\text{MS}} 1.90 \rightarrow 1.24$), while $|K|$ grows inward and the lapse pinches to zero at the centre.

The shrinking is physical; the regrowth that follows it

in the scans is not. In spherical symmetry a massless phantom can only shrink a horizon: with the double-null metric $-2e^{-f} du dv + R^2 d\Omega^2$, $T_{uv} = 0$ and $T_{vv} = -(\partial_v \phi)^2$, the Misner–Sharp mass on a marginally trapped tube obeys $\partial_v m = 4\pi R^2 e^f (\partial_v \phi)^2 \partial_u R \leq 0$ and $\partial_u m = 0$, and the tube is timelike or null [65, 75], so $R = 2m$ can only fall, as phantom accretion requires [66, 86]. In the three arms kicked with $\epsilon = +10^{-2}$ —at level 3, and at level 4 dressed with $\epsilon_2 = 0.005$ and 0.05 —the horizon falls by 40% in areal radius to a floor at $t \approx 47$ ($R = 2.34$), and the scans then show it growing back by 9–11%. That regrowth is not a measurement. It tracks a Hamiltonian-constraint violation that reaches the horizon at the floor time, it is the same in a purely spherical level-4 arm, and levels 3 and 4 agree on the horizon’s radius to a median 0.08% over the record, so it is not truncation error but most plausibly the evolved defect of our unconstrained seed (Sec. II F). The pure-quadrupole arm, whose seed has no $\ell = 0$ defect, only shrinks (Fig. 3(a)). What the collapse branch establishes is the shrink: the throat becomes a black hole that swallows its own phantom support and loses 40% of its horizon radius doing so, in minutes even at $10^6 M_\odot$ (Table IV). Hawking’s area law assumes the null energy condition [34, 70], which the phantom violates; this horizon runs it backwards.

F. The inflation branch

The inflation branch forms no trapped surface. The $\epsilon = -10^{-2}$ throat, evolved on an octant of an $L = 512$ box with finest $\Delta x = 1/16$ (the resolution of our $L = 64$ level-3 arms; Fig. 3), keeps growing to the end of the

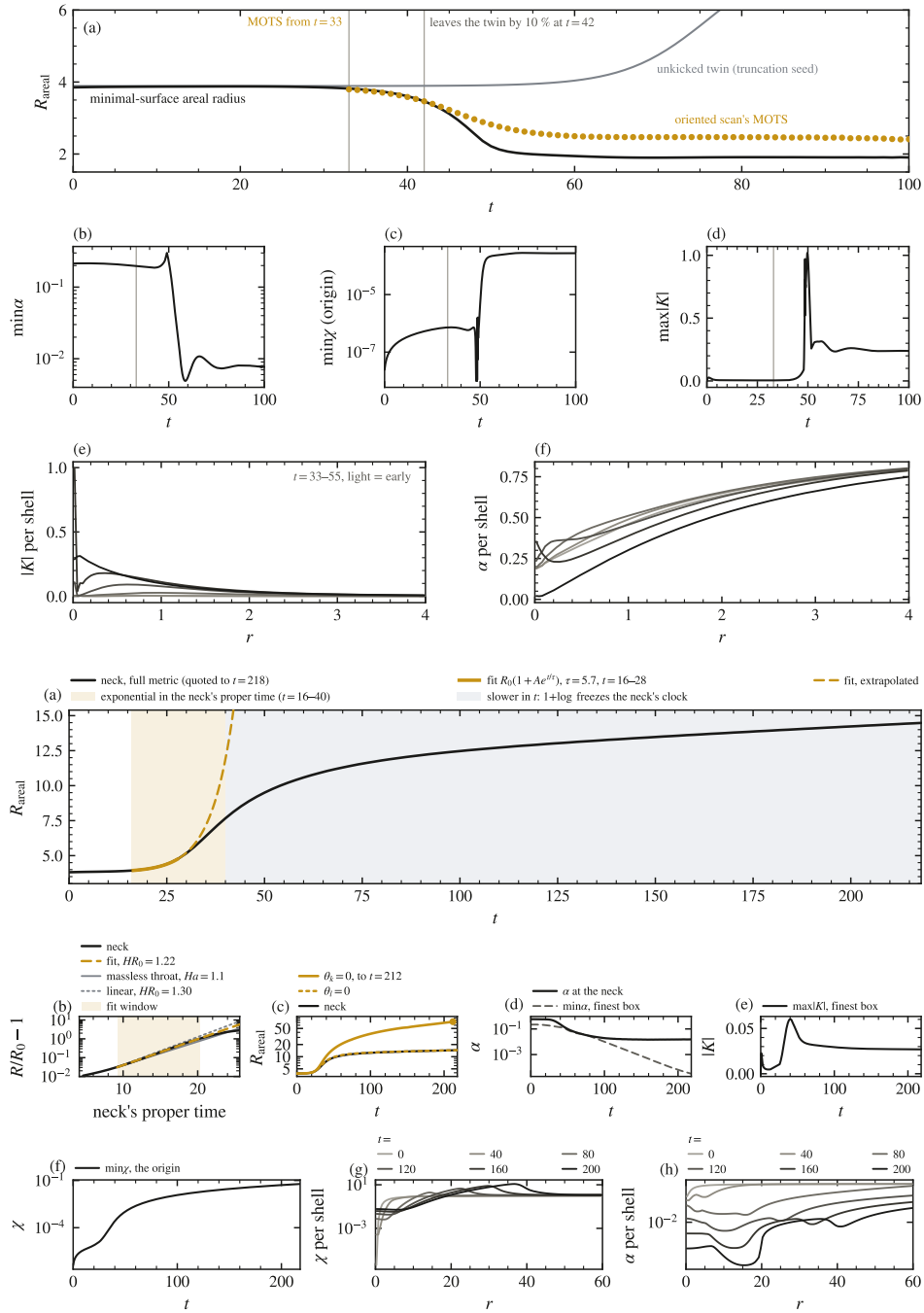


FIG. 3. *Top (collapse branch)*: Collapse of the pure-quadrupole arm ($\varepsilon_2 = 0.01$, no radial kick). (a) Minimal-surface areal radius (ink) and the scan's MOTS (gold) from $t = 33$: R 3.80 $\rightarrow$ 2.41, M_{MS} 1.90 $\rightarrow$ 1.24. Grey: the unkicked level-4 twin, which inflates on its own seed; this arm leaves it by 10% only at $t = 42$ —the second-order drive of the pure quadrupole. (b)–(d) Global $\min \alpha$, $\min \chi$ (the origin, the far side's compactified infinity) and $\max |K|$; the violent phase spans $t = 45$ –55. (e), (f) Radial profiles over 128 shells, $t = 33$ –55 (light = early): $|K|$ grows inward while α pinches to zero at the centre. Constraint norms: Fig. 10(c). The collapse ends in a black hole that consumes its supporting field. *Bottom (inflation branch)*: The inflating throat ($\varepsilon = -10^{-2}$; $L = 512$ octant, finest $\Delta x = 1/16$) to $t = 218$. (a) Neck areal radius in the full metric, 3.81 $\rightarrow$ 14.5; gold, the exponential onset, $\tau = 5.7$ over $t = 16$ –28; shaded, exponential growth in the neck's proper time ($t = 16$ –40, gold) and the slower growth in t once $1 + \log$ freezes the neck's clock (grey). (b) Over that window the neck's proper-time rate is $HR_0 = 1.22$ (local 1.0–1.3), against the massless throat's $Ha = 1.1$ [13] and the linear mode's 1.30. (c) The $\theta_k = 0$ horizon stays on the neck; $\theta_k = 0$ runs out to $R \approx 73$ by $t = 212$. (d) The lapse at the neck falls 0.57 $\rightarrow$ 0.015; dashed, its minimum on the finest box. (e), (f) $\max |K|$ and $\min \chi$ on the finest box, around the origin. (g), (h) χ and α per shell (light = early): the χ peak moves out with the neck. Constraint norms: Fig. 10(d). The throat keeps growing, anti-trapped; its onset is at the linear rate.

record, $\times 3.8$ by $t = 218$, and stays anti-trapped [76]: the $\theta_l = 0$ horizon sits on the neck and $\theta_k = 0$ runs outward, to $R \approx 73$ by $t = 212$. Until $t \approx 40$ the neck grows exponentially in its own proper time with $HR_0 = 1.22$ (locally 1.0–1.3; $R_0 = 3.81$, the kicked neck), 6% below our linear mode’s 1.30, consistent with this arm’s level-3-class resolution. Spherical evolutions find the same proper-time exponential: $Ha \approx 1.1$ for the massless throat [13] and exponential growth for massive ones [14]. After that the $1 + \log$ slicing freezes the lapse at the neck ($0.57 \rightarrow 0.015$), only ≈ 5 units of its proper time pass by $t = 218$, and its growth per unit t slows—a property of the slicing, not of the throat.

Its end state is open, and the throat is shedding its own support: the monopole scalar flux through every sphere grows quasi-exponentially over the quotable window (Sec. VIII G). Three endings remain: saturation at a larger radius, disfavoured because static drainholes are unstable fixed points; exponential inflation in proper time, which spherical evolutions find [13, 14] and our onset matches; or a turnover into collapse once too little phantom support remains, which a flux still growing when its reservoir is spent would force. That reservoir now has a number: outside the static throat the field carries a Misner–Sharp deficit of $R_\star/2 - m = 0.94$. Its spending is only bracketed. Over the window in which the exterior slicing is still quiet, the arms of Sec. VIII G shed 0.03 through $R = 18$ —a few percent of the store—while on this arm the $1 + \log$ front reaches every sphere before the monopole grows, and the energy through the coordinate-60 sphere by $t = 218$ reads 4.1 kinematically but 0.04 with the geometric factor of Sec. VIII F applied—either side of the store. A downturn of $R_{\min}(t)$ followed by a trapped surface would decide the ending; none occurs by $t = 218$, the growth holds at every sample, and the $\theta_k = 0$ surface keeps moving outward, with no slowing over the last fifty units; its rate in areal radius per unit coordinate time, ≈ 0.33 , depends on the slicing and is not a propagation speed.

G. Lifetime and physical scale

At fixed m/a the solution is scale-free, and the measured e-fold is $\tau\alpha_{\text{th}} = 0.78\text{--}0.87 R_\star$ in proper time—one light-crossing of the throat. The lifetime is a logarithm of the seed: the horizon times of Sec. IV C set a decade’s cost at $14M \simeq \tau \ln 10$, and the unkicked throat’s places the level-3 truncation seed at $\varepsilon \gtrsim 2 \times 10^{-6}$. Table IV converts these to physical units. No static throat of astrophysical mass survives: a stellar-mass drainhole is gone in milliseconds, a supermassive one in minutes, and only at $10^9 M_\odot$ does the lifetime reach days. The logarithm makes this robust: ten further decades of quiet initial data extend a $30 M_\odot$ throat from 1.6 to 22 ms. The same law prices a traversal. Normal matter crossing the throat collapses it, as spherical evolutions find for a traveller [13]; if a traveller of mass m_t seeds the mode at $\varepsilon \sim m_t/M$,

TABLE IV. The drainhole clock in physical units: the production throat ($R_\star = 3.89M$) scaled by its ADM mass M . $\tau = 5.9M$ is the e-fold time (Sec. IV B); the last three columns are the measured times to a horizon for the two declared kicks, both outward ($\varepsilon > 0$, the collapsing sign; Fig. 2(c)), and an upper bound for truncation noise (Sec. IV C). Intermediate seeds follow $t_{\text{BH}} \simeq 11M + \tau \ln(10^{-2}/\varepsilon)$.

M	$R_\star$	τ	time to a horizon		
			$\varepsilon = +10^{-2}$	$+10^{-3}$	noise ($\leq$)
$30 M_\odot$	172 km	0.87 ms	1.6 ms	3.7 ms	9.0 ms
$10^3 M_\odot$	5.7×10^3 km	29 ms	54 ms	0.12 s	0.30 s
$10^6 M_\odot$	5.7×10^6 km	29 s	54 s	2.1 min	5.0 min
$10^9 M_\odot$	38 AU	8.1 h	15 h	1.4 d	3.5 d

the throat stays open for $t_{\text{BH}} \simeq 11M + \tau \ln(10^{-2}M/m_t)$, which for $m_t = 100$ kg is $450M$, about 37 minutes, at $10^6 M_\odot$ and 57 ms at $30 M_\odot$. A foam-born pair must collide as it is born; otherwise each throat collapses or inflates on its own first (Sec. X A).

H. A type-I critical solution

In the language of critical collapse the static throat is a type-I critical solution [96]: a static solution with a single unstable mode (Sec. IV B), sitting on the threshold between two end states—collapse and inflation, where the standard problem has collapse and dispersal. The measurements above carry both type-I signatures. The first is time scaling: data a distance ε from threshold stay near the critical solution for a time $\tau \ln(1/\varepsilon)$ set by its one e-fold, which is the lifetime law of Table IV, $t_{\text{BH}} \simeq 11M + \tau \ln(10^{-2}/\varepsilon)$, and both branches leave along the same mode, each $\pm\varepsilon$ pair crossing at the same $t_\times$ (Sec. IV C). The second is a mass gap: black-hole formation turns on at finite mass, as at the static Bartnik–McKinnon solution of Yang–Mills collapse [97], not at the vanishing mass of type-II scaling. The $\varepsilon = +10^{-2}$ and $+10^{-3}$ horizons form at $R = 3.88$ and 3.81 , both close to the throat’s own $R_\star$: a tenfold smaller kick leaves the black hole’s mass at formation almost unchanged, where type-II scaling would shrink it as a power of ε . Shinkai and Hayward found evidence, in spherical evolutions of the massless throat, for critical solutions with a definite black-hole mass and for a collapse time tied to the input energy by an apparently universal exponent [13]. The measurements here identify the critical solution as the static throat itself, whose one unstable mode is measured in 3+1 dimensions and approaches the linear rate with resolution (Sec. IV B).

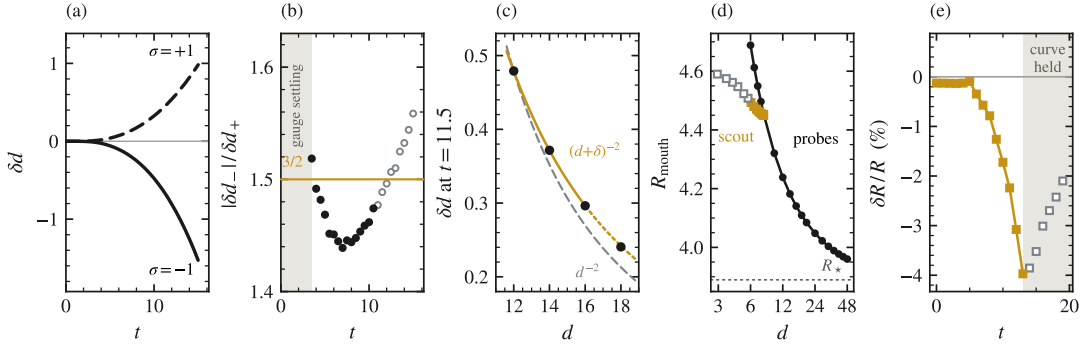


FIG. 4. Two throats at rest: the sign rule (a,b) and the force law (c) on the far-side-matched pairs (Sec. IID), and the placement calibration of the superposed campaign (d,e). (a) Separation change of two matched $d = 12$ pairs one sign apart: like-signed ($\sigma = +1$, dashed) opens, opposite-signed ($\sigma = -1$, solid) closes. Separations are inverse- χ -weighted centroids of the midplane χ pits. (b) Pull-to-push ratio against the prediction $(Q+1)/(Q-1) = 3/2$ (gold), where $Q = |F_\phi/F_{\text{grav}}| = (a^2 + m^2)/m^2 = 5$ is the scalar-to-gravity force ratio of Eq. (6): measured 1.462 ± 0.022 (scatter over 15 slices); grey, the gauge-settling window $t < 3.5$, excluded; open circles past $t = 10.5$, where the closing pair has eaten a tenth of its gap and the displacements are no longer small, excluded too. (c) The force law on the matched ladder: how far like-signed pairs released from rest at $d = 12/14/16/18$ have pushed themselves apart by $t = 11.5$ (dots), against inverse-square in the coordinate separation scaled through $d = 12$ (grey, dashed) and $A/(d+\delta)^2$ fitted on $d = 12-16$ alone (gold), dotted where it predicts the $d = 18$ rung (Sec. VB). If the push were inverse-square in d , every dot would lie on the grey curve; the dots rise above it, and the gold fit absorbs the excess with the offset δ . (d) Per-mouth areal radius of freshly placed pairs at $t = 0$ (circles), exceeding the isolated R_* (dotted) by an excess $\propto d^{-1.16}$. Gold squares: the head-on scout's mouths at the separation reached; open grey squares (here and in e) lie closer than the closest probe. (e) The scout against the placement curve, $\delta R/R = R_{\text{scout}}/R_{\text{place}} - 1$: the -0.13% plateau to $t = 5$ calibrates the tracker's grid snap, and the squeeze then grows to -4.0% by $t = 13$. In the shaded band the curve is held at its last point, so the magnitude is a lower bound; a boost can only raise a coordinate-sphere reading, so a reading below the curve is a genuine squeeze.

V. TWO THROATS AT REST

A. The sign rule

Two identical throats released from rest push apart; reversing one throat's scalar, $\sigma = +1 \rightarrow -1$ in Eq. (6), turns the push into a pull (Fig. 4(a,b)). Both forces share one radial law, so their ratio Q is separation-independent: no gap, mass or approach speed lets a like-signed pair fall together, and $\sigma = -1$ is the only gravity-driven merger channel. The flip is not a tuning: $\phi \rightarrow -\phi$ leaves T_{ab} , and with it the one-body geometry, unchanged. The measured pull-to-push ratio at $a = 2$ is 1.518 ± 0.021 (scatter over 15 slices) against the $(Q+1)/(Q-1) = 3/2$ of Eq. (6), and the force follows the throat width—the scalar charge—not the mass, which is equal in all arms. Rerun on matched data (Sec. IID), the same pairs give 1.462 ± 0.022 : the rule and its ratio belong to the throats, not to the superposed placement.

B. The force law

How the push falls with distance is read off a ladder (Fig. 4(c)). Four like-signed pairs, identical but for the gap $d = 12/14/16/18$, are released from rest, and by a common $t = 11.5$ their separations have grown by $\delta d = 0.4696/0.3699/0.2980/0.2438$. For displacements

this small, δd at a fixed time is proportional to the force, so under the inverse-square law of Eq. (6) $\delta d d^2$ would be the same on every rung. It is not: it rises monotonically, $67.6 \rightarrow 79.0$, across the ladder, which excludes inverse-square in the coordinate separation. Instead $F \propto (d+\delta)^{-2}$ with $\delta \approx 3-4$ describes it: δ fitted on $d = 12-16$ alone predicts 0.246 at $d = 18$, where inverse-square gives 0.209 and the measurement is 0.2438 . Since δ is comparable to the placed pair's mouth radius (4.24 at $d = 12$; Sec. VC), the force is inverse-square in an *effective* separation exceeding the coordinate one by about a throat radius. The matched rerun keeps the ladder—no rung moves by more than 2.0% at $t = 11.5$ —and the full-ladder fit tightens to $\delta = 2.65$ against the superposed 3.56 , the direction the smaller matched mouths set. Across widths, at $d = 12$, the push grows at every rung ($\delta d = 0.147/0.283/0.416/0.615$ by $t = 11$ for $a = 1/1.5/2/3$) with an effective exponent of $1.3-1.4$; the point-charge a^2 scaling is not confirmed.

C. Placement systematics

Freshly placed pairs read $+1.8\%$ ($d = 48$) to $+20.5\%$ ($d = 6$) wider than the isolated throat, falling as $d^{-1.16}$ (Fig. 4(d,e)), so every same- d comparison here is made against that curve rather than against the lone throat. Read that way, the released head-on pair's mouths are genuinely squeezed, -0.6% at $t = 7$ growing to -4.0%

by $t = 13$ —an interaction effect, since the lone throat holds its radius to 0.1% until $t \approx 35$. The excess is the superposition’s rescaling of each throat, not a measurement artefact: the mouths’ far-side masses carry it too (Sec. IID), and the matched placement removes it.

VI. THE HEAD-ON MERGER

A. Contact and the common horizon

Released from rest— $p = 0$ —at $d = 8$ with opposite scalar signs, on matched, solved data ($M_{\text{ADM}} = 2.3573$), the pair falls together, and the spectral finder (Sec. IIID) holds a common MOTS around both mouths from $t = 18$ (Fig. 5): born while the χ pits are still ~ 3.4 coordinate units apart, a tenth out of round (deformation 0.104), with $R = 5.634$, $M_{\text{MS}} = 2.817$ and 1.05 times the two throats’ summed area. No surface ever closes about either mouth alone: the horizon is born common. At birth the Penrose inequality is violated outright— $R/2 = M_{\text{MS}}$ exceeds M_{ADM} by 19.5%—as the phantom permits, since the inequality presupposes the non-negative energy density it lacks [79, 80]: each throat’s own Misner–Sharp mass, $R_*/2 = 1.94$, already exceeds its ADM mass of 1 by the negative energy of the field outside it. From birth the horizon only shrinks (Sec. VIB): Hawking’s area theorem run backwards, as the first law demands of an influx with $T_{ab}k^ak^b < 0$. Formation is robust to the seed: in the superposed campaign an ensemble carrying $\varepsilon = \pm 10^{-2}$ on both throats—the amplitude that decides a lone throat’s fate—found the same surface at $t = 22.0$, areal radius within 1.1% and mass within 0.5%. The seed sets a lone throat’s fate and the grid’s lifetime, not the pair’s. The round star scan, the live monitor, finds the birth 4 units late and 11% small and goes blind while the surface is deformed (Fig. 5(a)): the measured cost of scanning a deformed surface with round apertures, and Sec. IIID’s reason to read every horizon number here from the finder.

B. Past the interior failure to a black hole

The merged core is the grid’s hard part, not the space-time’s. Twenty-one units after the horizon forms, the level-5 leg dies of a NaN in the metric component h_{11} in a single cell at the merged core ($t = 38.845$)—inside the MOTS, with the global constraint norms clean to the last step: the censored interior failure of Sec. VIID. A level-6 leg restarted from $t = 35$ walks through the failure window, and a level-4 leg carries the remnant from $t = 50.8$ to $t = 100$ with no NaN and no interior device. The finder’s history is continuous across both seams (R steps by 0.08% and 0.04%), and in-code Ψ_4 is identical on the overlaps: the seams carry nothing.

The remnant is a black hole with both throats inside it, and it shrinks as it swallows the phantom field that held them open (Fig. 5(e)): R falls 15.2% from birth—28% of

the birth area gone—to $R = 4.7787$ at $t = 100$, rounding from deformation 0.104 to 0.004, while M_{MS} falls from 2.817 to 2.3894. Gravitational waves carry only 1.7% of that loss (the measured $E_{\text{GW}} = 3.1 \times 10^{-3} M$, Sec. VIIB); the rest is the negative-energy field going down the hole. The shrink is monotone on the finder’s 84 rows—no row rises on its predecessor by more than 0.08%, and unlike the single throat’s star scans (Sec. IVE) the history never grows back—and it can only be: on a marginally trapped tube a canonical field can only grow a horizon, and the phantom can only shrink it, the spherical first law of Sec. IVE. That law is measured on this surface. Over $t = 51$ –60, replayed with every plotfile kept, the observed $\Delta R = -4.1 \times 10^{-3}$ is bracketed by the law evaluated on the found surface: -7.0×10^{-3} from the phantom influx alone, -2.5×10^{-3} with the outgoing shear’s growth term added. By $t = 90$ –100 the influx is spent: the phantom rate on the surface sits 69-fold below its $t = 45$ –55 value, the total first-law rate 112-fold, and the radius drifts at 3.2×10^{-4} per unit. The late decline is exhaustion, not accretion.

The end state is the Schwarzschild sphere of the pair’s Bondi mass. At $t = 100$ the horizon sits 1.4% above $2M_{\text{ADM}}$ in radius and 1.4% above M_{ADM} in mass: with $R = 2(M_{\text{ADM}} - E_{\text{GW}} - E_\phi)$ and the measured E_{GW} , the offset puts $E_\phi = -0.039$, negative as the ghost channel radiates and of the sign and order of the flux integrals of Sec. VIIG. The $d = 6$ orbital merger’s remnant settles within 2.2% of the same radius (Sec. VIIC), as it should if the end state is set by the pair’s mass.

C. The half-mass pair

Halving the throat mass moves everything earlier without avoiding the horizon. At $m = 0.5$, $d = 6$, the lapse falls from 0.44 to 0.038 over $t = 7.5$ –13, the tracker merges the mouths at $t = 11.1$, the offline scan finds a common MOTS at $t = 12$ ($R = 3.83$), and the run dies at $t = 14.0$ with the level-3 scout’s anatomy (Sec. VIB): χ on its floor, $\max |K|$ running away and a one-step overflow of the metric inside the common MOTS. The death belongs to the pair: the lone $m = 0.5$ throat is exact to 0.02% through $t = 14$ and fails only at $t = 26.3$, and differently—in K , at its compactified origin, the origin failure of Sec. IVA.

VII. THE ORBITAL CHANNEL

A. Merger or fly-by

Varying the tangential momentum p at $d = 12$ on boosted, solved data separates scattering from plunge (Fig. 12 in Appendix B). The two slower arms scatter: at $p = 0.25$ the pair passes at 2.32 ($t \approx 48$) and recedes to 8.5 by $t = 100$; at $p = 0.45$ it passes closer, 2.18 at $t \approx 48$, and recedes to 5.3. Both reach their stop

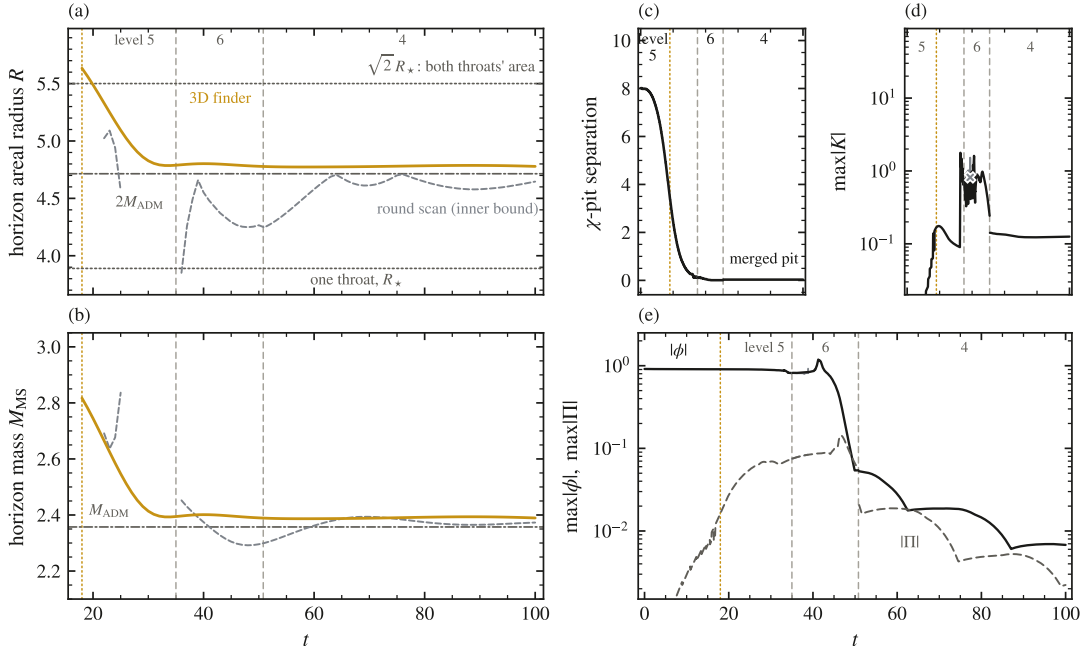


FIG. 5. The head-on merger ($d = 8$ from rest, $p = 0$, opposite signs) on matched, solved data ($M_{\text{ADM}} = 2.3573$), and the black hole it makes. One evolution, drawn across its three legs (faint dashed rules at the seams, $t = 35$ and 50.8 ; levels named in each panel): the level-5 leg reaches contact and then dies of a NaN in the metric component h_{11} at the merged core at $t = 38.845$ —inside the MOTS, the censored interior failure of Sec. VIID, its overrun drawn grey to the death cross in (c)–(e); the level-6 leg restarts from leg 1’s $t = 35$ checkpoint and walks through the wall; the level-4 leg carries the remnant to $t = 100$ with no NaN, the fine levels no longer needed once the wall is crossed. In-code Ψ_4 is identical on both overlaps: the seams carry nothing. (a) Areal radius and (b) Misner–Sharp mass of the common MOTS. Gold: the spectral finder’s history (Sec. IIID, one row per plotfile, the three replay legs joined), born at $t = 18$ with $R = 5.634$ ($M_{\text{MS}} = 2.817$) and settling—monotone to 0.08% per row—onto the Schwarzschild sphere of the pair’s mass: $R = 4.7787$ and $M_{\text{MS}} = 2.3894$ at $t = 100$, 1.4% above $2M_{\text{ADM}}$ and 1.4% above M_{ADM} . Grey dashes: the round star scan, the demoted inner bound (Sec. IIID)—it finds the birth 4 units late and 11% small, goes blind over $t = 26$ – 35 while the surface is deformed, and wobbles about the settle; its rows are joined only within a contiguous run, never across the gap. Rules: one isolated throat, $R_* = 3.8895$; $\sqrt{2}R_*$, a sphere with both throats’ area; $2M_{\text{ADM}}$ (M_{ADM} in (b)). (c) Separation of the χ pits. (d) $\max|K|$. (e) Grid maxima of $|\phi|$ (solid) and $|\text{II}|$ (dashed): the field is swallowed. Gold dotted rule: horizon birth.

times with no NaN—the $p = 0.45$ arm at stock settings to $t = 100$, so the channel has no wall of its own below the plunge boundary—and the finder sees no trapped surface on any plotfile of either. The two faster arms plunge: $p = 0.60$ reaches contact at $t \approx 40$ (pit separation 1.86 at its first leg’s stop) and $p = 0.90$ just past its $t = 40$ leg (2.4 and still falling), each collapsing to a merged core (Sec. VIID). The plunge boundary therefore sits between $p = 0.45$ and 0.60 : 90 – 120% of the circular momentum under the combined pull, bracketing the circular value itself. That pull, $1 + Q$ times gravity’s (Eq. (6)), is central: a pair that misses contact keeps its angular momentum and swings past. But the approach deepens with speed—closest approach falls from 2.32 at $p = 0.25$ through 2.18 at 0.45 to contact at 0.60 , against the Newtonian expectation of a pericentre growing with $J = pd$ —so capture here is contact, not an innermost orbit. Nor do the scatterers escape: both start below the circular momentum and remain bound in Newtonian terms; the vacuum innermost stable circular orbit does not apply to a pair bound by a pull six times gravity’s, and Sec. XA shows

the throats’ own decay would end any wider orbit within a few periods. What follows a pass is set by the throats, which inflate (Sec. VII B).

The $p = 0.25$ fly-by ($L = 128$, level 5 to $t = 100$) is the production scattering pass. The turn at closest approach is orbital, but the recession after it is not that of two compact throats: both mouths inflate, the constraint norm crosses the campaign’s 2.5×10^{-2} criterion at $t = 63.3$ as they do, and nothing is quoted from the run after that time, nor from its waves after the matching retarded time (Sec. VIID). By then the encounter is an expanding cloud of the field that supported the throats, not two receding wormholes.

B. The race between contact and inflation

At $d = 12$ no arm merges at any momentum. The mode-3 spiral with Bowen–York momentum (the systematics arm), its exact-boost rerun, and an inward-momentum scout ($p = 0.25$ aimed straight at the com-

panion) all end the same way: the mouths inflate faster than the orbit closes—the scout’s $R_{\min}$ grows +4.7% by $t = 8$ and +72% by $t = 24$ while its infall stalls near separation 2.2—and the finder holds no common MOTS on any plotfile of the three (the boosted arm dies of the interior failure at $t = 71.78$ having never trapped within its search window).

The mechanism is the branch selection of Sec. IV C, run by the companion. A lone throat’s fate is opposite to a declared kick: pushed outward it collapses, squeezed it inflates. A companion squeezes: solving the constraints with the second throat present compresses each mouth, by 0.4% at $d = 12$ and 1.2% at $d = 8$, the same in superposed and solved data—the physical interaction, not a data defect—so every pair’s mouths start on the inflation branch, at any momentum, and grow on the throat’s own e-fold time $\tau \approx 5.88$ (Sec. IV B). That the solve keeps a declared kick is checked directly: the $\varepsilon = +0.1$ kicked single, rerun through the constraint solve, is born trapped (a MOTS on its $t = 0$ slice, $R = 4.306$), its superposed twin’s fate. A merger is therefore a race against the inflation it starts: the head-on’s horizon arrives 3.1 e-folds in and wins; the $d = 6$ orbit’s arrives at 2.2 e-folds and wins (Sec. VII C); the $d = 12$ approach, 8 e-folds to its closest pass, always loses. The design rule for an orbital merger follows: short separation and small tangential momentum.

C. The $d = 6$ merger

The design point that wins the race is $d = 6$ with a small tangential twist, $p = 0.10$ per mouth. A level-3 scout merges, and the production chain repeats it on the $L = 128$ box (Fig. 6): the spectral finder holds a common MOTS around both mouths from $t = 13$ —2.2 e-folds into the inflation, the pits still 3.6 apart—born with $R = 5.60$ ($M_{\text{MS}} = 2.80$, deformation 0.115) and shrinking monotonically ($\leq 0.05\%$ per row over the finder’s 88 rows) by 12.8% to $R = 4.883$, $M_{\text{MS}} = 2.441$, deformation 0.016 at $t = 100$: a settled black hole within 2.2% of the head-on remnant’s radius, reached through an orbit. As in the head-on, the horizon is born common—no surface ever closes about either mouth—and it swallows the pair mid-inflation.

The chain is the head-on’s pattern stretched over a harder wall (Sec. VII D): level 5 from $t = 0$, stopped at $t = 26.26$ as the merged core’s K ran away; a $\sigma_{\text{KO}} = 1.0$ leg through the K wall to $t = 30.5$; the χ -floor leg (min χ raised to 10^{-4}), which sails through the single-cell wall, past every earlier death, to $t = 36.6$; and a level-4 settle leg, the wall knobs reverted, to $t = 100$ with no NaN—the cured interior running at its own $\chi_{\min} \sim 7 \times 10^{-5}$, below the crossing leg’s floor, with no cap needed. The floor is a crossing device, confined to $t = 30$ –36.6 inside the horizon; everything quoted here comes from the finder’s surface outside it.

D. Interior failure after horizon formation

Two failure modes stand between a merged core and its settled remnant, and each now has a measured cure. The first is a single-cell blow-up of the metric component h_{11} at the merged core’s centre: the head-on’s level-5 leg at $t = 38.845$, the $d = 6$ chain’s restarts at $t = 27.5$ –30.5, the $p = 0.60$ plunge at $t = 55.5$ —in every case with the global constraint norms clean to the last step. It is not under-resolution: on the $d = 6$ chain the level-6 restart dies at $t = 29.4$ and the level-7 restart dies *earlier*, at $t = 27.5$ with $\max |K|$ still climbing—a finer grid tracks the core’s collapse further and meets the failure sooner—and raising the dissipation to $\sigma_{\text{KO}} = 1.0$ delays the cell by a unit without curing it. What cures it is the χ floor: with $\min \chi$ capped at 10^{-4} the $1/\chi$ steepness runaway cannot complete, and both the $d = 6$ chain and the $p = 0.60$ plunge cross their death points and run to $t = 100$, the $d = 6$ settle leg then running floor-free. The second mode is the K wall: $\max |K|$ itself running away at the merging core, $93 \rightarrow 172$ in one step on the $p = 0.90$ arm at $t = 45.3$ with χ nowhere near its floor ($\chi_{\min} = 5.4 \times 10^{-3}$)—the floor does not touch it. Its knob is resolution: the $d = 6$ chain’s level-6 restart cleared the same runaway that stopped its level-5 leg.

Both failures are censored. The core fails only after a common MOTS encloses it: the $d = 6$ cell NaNs arrive 15–18 units after its horizon, the head-on’s 21 after its, and lone collapses die behind their horizons at level 3 too ($\varepsilon = +10^{-3}$ at $t = 40.1$, 15 units after its horizon forms; $\varepsilon = +10^{-2}$ dressed with $\varepsilon_2 = 10^{-2}$ at $t = 27.6$, 14 after), while no level-4 lone collapse dies. The failure occupies one cell (the K wall, a few), the exterior is already set, and the finder’s surface rides through every crossing unmoved: whatever diverges is the coordinate representation of a merged interior, behind the horizon.

The same anatomy was mapped exhaustively on the superposed campaign, kept as the systematics study. Restarted from one $t = 50$ checkpoint at maximum levels 3–7, every arm of the superposed $p = 0.12$ spiral died on the h_{11} NaN, with the gains per doubling saturating: +1.03, +2.51, +0.53, +0.06 units (Fig. 11(a) in Appendix A). Sixteen times the resolution bought 4.13 units, halving the time step on level 5 bought +0.36, the core matter damping under a unit. Nor was the failure a stronger core at higher momentum, as far as $\max |K|$ can tell: across the superposed momentum scan the level-3 failure arrived at $t = 52.1$ –53.4 with no trend in p ; no level-5 arm showed a runaway in $\max |K|$ before its NaN, the $p = 0.15$ –0.25 arms dying at $\max |K| = 0.71$ –1.05; and the head-on passed a peak of 1.36 and survived.

The natural suspect is the slicing. Every arm uses 1+ log slicing with a Gamma-driver shift, whose slicings can develop coordinate shocks [54] that refinement resolves rather than cures; a shock at a young, strongly deformed horizon would end every grid. Four gauge choices move the clock without removing the failure (Fig. 11(b); the three non-standard arms also carry the core matter

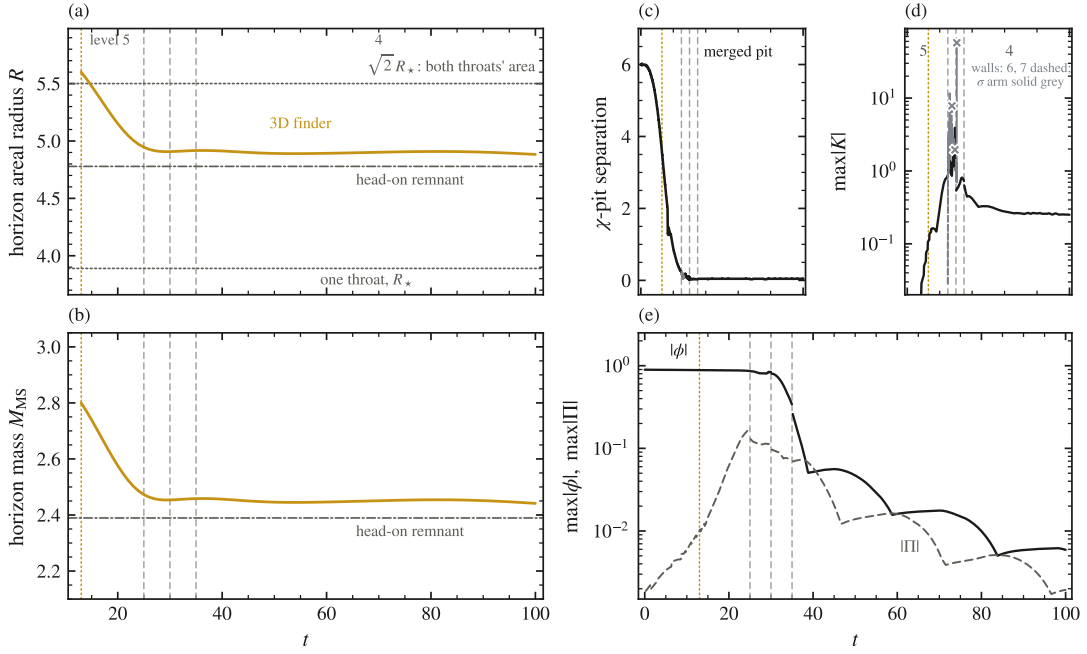


FIG. 6. The orbital merger ($d = 6$, tangential $p = 0.10$, opposite signs, boosted mode-3 data) carried through its merger wall, and the black hole it settles to. One evolution composed across its legs (faint dashed rules at the restarts, $t = 25, 30$ and 35 ; levels named in the panels): level 5 from $t = 0$, stopped at $t = 26.26$ as the merged core's K ran away; the $\chi_{\text{KO}} = 1.0$ leg; the χ -floor crossing leg; the level-4 settle leg to $t = 100$, no NaN. (a) Areal radius and (b) Misner–Sharp mass of the common MOTs: gold, the spectral finder's history (Sec. III D), born at $t = 13$ and settling from above onto the head-on remnant's values (dash-dotted rules, $R = 4.7787$; $M_{\text{MS}} = 2.3894$ in (b)). Dotted rules in (a): one isolated throat, R_* ; $\sqrt{2}R_*$, both throats' area. (c) Separation of the χ pits (grey: leg 1's overrun past its restart checkpoint). (d) The wall: $\max|K|$ of every arm—the used legs in ink; the level-6 and level-7 diagnostic restarts (grey dashed) die at $t = 29.4$ and, *earlier*, 27.5 , each a single-cell h_{11} NaN at the core's centre with the global norms clean to the last step; the σ arm (grey solid) clears the K wall and dies at the same cell at $t = 30.5$ (death crosses); the χ -floor and settle legs carry the history through. (e) Grid maxima of $|\phi|$ (solid) and $|\text{II}|$ (dashed): the field is swallowed. Gold dotted rule: horizon birth.

damping, which alone moves the level-3 failure by under a unit). At level 3 the spiral dies at $t = 43.6$ with the $1+\log$ coefficient halved, 49.0 with the harmonic-class driver $-2\alpha^2 K$, 52.1 in the standard gauge, and 61.9 with the shift damping raised $\eta = 1 \rightarrow 4$, which delays the whole sequence by ~ 10 units (the tracker merging the two centres at $t = 41.7$ against 32.1, χ floor at 55.6 against 44.9): the level-3 clock gains because the whole sequence is late—its failure still follows its χ floor by 6.3 units against 7.2—not because level 3 resolves more. On the head-on control the same knobs give deaths at 25.2 (harmonic), 26.9 (standard) and 34.2 ($\eta = 4$), and trapping precedes each: the harmonic arm's MOTs appears at $t = 20.8$ with $R = 5.47$ against the standard 5.40—the same surface on an earlier clock, 4.4 units ahead of the failure—and the $\eta = 4$ arm carries a common MOTs from $t \leq 30.0$ ($R = 5.41$), which a level-5 restart carries past the level-3 failure to $t = 40$. In the spiral under $\eta = 4$ the common MOTs is present at level 5 from $t \leq 55.2$ ($R = 4.83$, M_{MS} within 0.3% of the standard gauge's) and at level 3 at $t = 60.01$ and 61.5 , before that arm's failure at 61.92. Under the harmonic-class driver the spiral shows only a $\theta_{\text{out}} = 0$ surface that is not trapped,

0.03 before the failure, so censorship there remains open. Refinement narrows the gauge spread without closing it: the $\eta = 4$ arm restarted at level 5 from $t = 50$ dies at 60.04, 1.9 units *before* its level-3 clock (a restart at $t = 60$ fails at once, at 60.05), while the standard gauge on the same box and restart dies at 55.60—a separation of ~ 4 units against 9.8 at level 3.

Within the Bona–Massó slicings and shift dampings tried the failure always occurs, at a gauge-dependent time. The record thereby tests weak cosmic censorship outside the matter it is posed for: censorship is conjectured for physically reasonable matter [98, 99], and the phantom violates the null energy condition. In every collapse resolved here—the head-on, the spiral at level 5 in the standard and the $\eta = 4$ gauge, and the lone collapses—the evolution fails, if at all, only inside a MOTs; the harmonic-class spiral is the one case left open. We have not determined whether a curvature singularity reaches the slice at the failure, so this is evidence for censorship under null-energy violation, not a proof of it. Every exterior quantity we quote is taken before the failure or from the interior-freeze evolution, whose influence reaches the extraction spheres only after the burst

has passed (Sec. VII C).

VIII. GRAVITATIONAL WAVES

Every channel radiates, and at our extraction radii the most energetic is the fly-by, which forms no horizon while its throats lie inside the scan window. The peak $|r\Psi_4|$ of the dominant mode at the innermost sphere is 4.1×10^{-2} for the fly-by, 2.9×10^{-2} for the spiral, 2.4×10^{-2} for the head-on, 1.1×10^{-2} for the vacuum twin and 2.9×10^{-3} for the collapsing throat (Fig. 7). Each wormhole record propagates outward at c within its output cadence (Fig. 7, right); only the vacuum twin’s waveform lag exceeds its cadence, by 3.4 units across its two spheres, its inner sphere sitting in the near zone. No sphere is asymptotic: the dominant periods are comparable to the extraction radii, and the peak $r\Psi_4$ falls across the spheres by 22% on the head-on and 40% on the fly-by, whose outermost reading drops below the spiral’s. We therefore do not extrapolate, and carry the spread over spheres as the near-zone error bar of every energy (Fig. 8(d)).

The records are also clean of the initial data’s junk radiation. The seeded and superposed starts are not constraint-solved (Secs. II C and II F), but a defect radiating from $t = 0$ crosses an extraction sphere by $t \approx R$ —one light-crossing—while every quoted burst arrives with its physical trigger, tens of units later: the spiral’s merger transient at $t = 41.2$, the fly-by’s pass (all spheres peak at $t - R \simeq 34.5$), the head-on’s horizon at $t = 22$; the per-sphere envelopes rise with the burst, not at $t - R \approx 0$ (Fig. 7, right). The junk amplitude is measured where the defect is largest: the spherically kicked throat, whose shell defect peaks at $0.93 \times 16\pi|\rho|$ (Sec. II F), radiates nothing above the $\ell = 2$ numerical floor (Sec. VIII A), and the quadrupolar arms’ bursts are linear in ε_2 and unchanged when the radial kick is removed (Fig. 11)—the throat’s wave, not the shell’s transient. On the binaries the momentum constraint holds exactly at $t = 0$ and the Hamiltonian defect is $O(a^2/d^2) + O(m/d)$ (Sec. II C); its lasting effect is to excite the throats’ unstable mode, at an effective seed of 2×10^{-4} – 6×10^{-4} (Sec. X A).

In the dominant multipole the fly-by radiates $4.2 \times 10^{-2} M$ through the outermost sphere ($7.4 \times 10^{-2} M$ through the innermost, where near-zone content is largest) and the spiral up to $2.2 \times 10^{-2} M$. For scale, the closest non-merging black-hole scatterings computed in vacuum radiate $\approx 2 \times 10^{-2}$ of their total mass in all multipoles [81]; the drainhole pair is pulled six times harder. The spiral’s own vacuum control, a binary black hole with its d and p , radiates $2.6 \times 10^{-3} M$ (grey in Fig. 7(c)), $8.5\times$ less; the fly-by’s own control is a vacuum binary at $d = 12$, $p = 0.45$. In vacuum that momentum is unbound (circular 0.204, parabolic 0.289), so the black holes start at periapsis and coast apart, radiating $1.05 \times 10^{-3} M$ in a single cycle as they swing away (grey in Fig. 7(d): peak $|r\Psi_4| = 9.0 \times 10^{-3}$ at the fly-by’s sphere, $4.5\times$ below the fly-by’s), whereas the drainhole pair falls from 12 to 4.8

under a pull six times gravity’s. The fly-by thus outradiates a vacuum binary of its own parameters by 40 – $70\times$. The vacuum controls use the settings of `GRTEClyn`’s black-hole example (Sec. III B), so these ratios include a numerical systematic we have not quantified.

The closest positive-energy comparison is a boson-star binary, likewise horizonless and held up by a scalar field. Its mergers end in a boson star or a black hole [100–102], unless the pair is a boson–antiboson one, which all but annihilates [101], and head-on outcomes and wave output depend on the stars’ relative phase [103]. The negative-energy support changes how a horizonless encounter ends: the fly-by leaves no star to relax and ring down but two mouths that inflate (Sec. VII A), while the field that held them open radiates energy of the opposite sign (Sec. VIII F).

A. Collapsing throat

A lone throat radiates only if its collapse is not spherical. The spherically kicked arm emits nothing above the $\ell = 2$ numerical floor, while the same kick with an $\varepsilon_2 = 0.05$ quadrupole produces a burst of peak $|r\Psi_4| = 2.9 \times 10^{-3}$, $44\times$ the floor of the matched level-4 control at $R = 14$, carrying $2.6 \times 10^{-5} M$ (integrated to $t = 58$, before the numerical floor dominates). The burst arrives in radius order at 0.8 – $1.0 c$, carries the same $r\Psi_4$ to four spheres within 7.2%, and is linear in the seed (Fig. 11 in Appendix B): $4.85\times$ for a $5\times$ kick, ratios $1.99/1.97/1.87$ at $R = 10/14/18$ for a halved seed, linear to 1–16% across $\varepsilon_2 = 0.005$ – 0.05 . The pure quadrupole— $\varepsilon_2 = 0.01$ with the radial kick removed—radiates the same amplitude to 2%, its waveform leading the kicked arm’s by 1.1 – $1.5 M$: the deformation sets the wave, the kick only its clock. These seed amplitudes follow the wave, not a decade grid: $\varepsilon_2 = 0.01$ matches the radial kick, whose usable range ends at 10^{-2} (Sec. IV C); 0.05 was added when the 0.01 burst stood only $\approx 2\times$ above the spherical control’s floor, too little to read a ringdown; and 0.005 halves the headline seed. A 10^{-3} arm would sit below that floor (Fig. 11(b)), and a 10^{-1} arm would deform the throat’s poles by $\approx 20\%$, as much as the $\varepsilon = \pm 0.1$ radial seeds that do not survive. The remnant rings at period $20.6 M$ (dashed in Fig. 7(a); an independent least-squares fit gives 20.61 ± 0.15), with no drift across the window (half-periods 9.4 , 10.3 and $9.0 M$), consistent with the Schwarzschild fundamental of a black hole of mass 1.16 – 1.21 (period 19.6 – $20.4 M$ [83]), close to the horizon mass near its minimum. After that minimum our horizon masses are not reliable (Sec. IV E), so we claim no more than this consistency. It does not follow the larger mass the horizon carries at the window’s start (1.47 , whose fundamental period is $24.7 M$). The decay time is not measurable on this record.

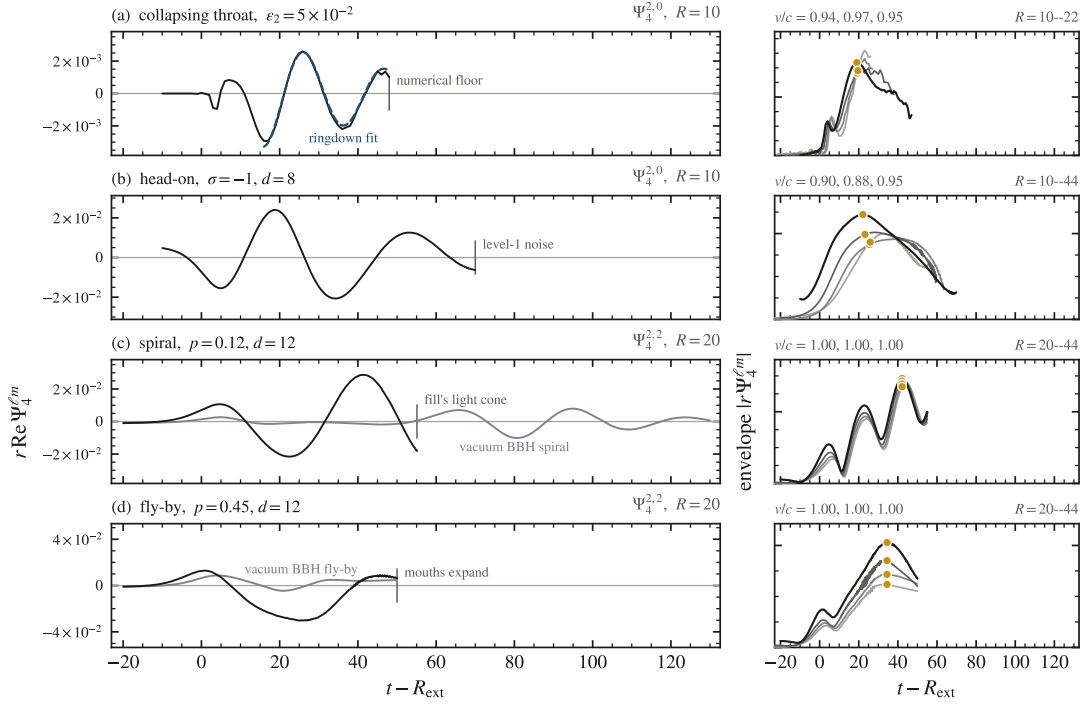


FIG. 7. The waveform gallery: one row per wormhole scenario on one retarded clock. Left: $r \text{Re} \Psi_4$ of the dominant mode at the innermost sphere, each row on its own scale; row (a) is the $\varepsilon_2 = 0.05$ arm, the only one loud and long enough to carry the ringdown, with the damped-sinusoid ringdown fit of Sec. VIII A dashed (period $20.6 M$). Caps mark where the drawn record ends: the collapsing throat where its numerical floor, the spherical control's own $r \Psi_4^{2,0}$, passes 10% of the burst peak; the spiral, carried from $t = 57$ by the frozen-core arm, at that core's light cone $t = 57 + (R - 1.9)$; the fly-by, complete to $t = 100$, at $t - R = 50$ on every sphere (Sec. VIII D); the head-on, drawn from its production chain's glued record (the caption of Fig. 5), where the late-time noise of its level-1 refinement cube (from $t = 80$ at $R \leq 20$; level-0 spheres run to the end) reaches each sphere. Grey in (c) and (d): the vacuum BBH spiral and fly-by, black holes with the same d and p (Sec. VIII), at the same sphere and on the same scale. The vacuum spiral's merger comes later and lower than the drainhole burst (Sec. VIII E); the vacuum fly-by, drawn over the fly-by's window, radiates one cycle as the holes swing away from periapsis and nothing at the pass, where the fly-by radiates its arch. Right: the amplitude envelope at every sphere on its own retarded time (ink to faint, inner to outer): $|r \Psi_4|$ for the complex $(2, 2)$ modes, the analytic-signal envelope for the real $(2, 0)$ modes, whose $|r \Psi_4|$ would fall to zero at each node. A signal moving at c and falling as $1/r$ folds onto one curve; each envelope rises with its burst (collapse, plunge, pass) and stops before a contaminant grows into it. Gold dots: the correlated wavefront. The quoted v/c is the lag of the whole complex waveform between neighbouring spheres over the drawn records, which, unlike peak timing, survives flat-topped envelopes (Sec. VIII E). Speeds: throat 0.94/0.97/0.95, head-on 0.94/0.88/0.95, spiral 1.00/1.00/1.00, fly-by 1.00/1.00/1.00.

B. Head-on ringdown

The head-on's signal is a bell with no chirp: the $(2, 0)$ breathing mode at $R = 10$ swings through $+0.023, -0.018, +0.011, -0.006$ at $t = 28.2, 43.8, 63.4, 82.2$ —a period of ~ 31 – 39 , the amplitude falling $\times 0.6$ – 0.8 per half-swing—and arrives ~ 4 units later per 4 units of extraction radius ($0.94 c$), outgoing across all three spheres. On the detector track it is the bell that slows from 707 to 410 Hz in Fig. 8(c).

C. Spiral burst

The spiral's $(2, 2)$ burst peaks at $|r \Psi_4| = 2.9 \times 10^{-2}$ and folds onto one curve across $R = 20$ – 44 at $v/c = 1.00/1.00/1.00$. The burst is launched by the merger tran-

sient at $t = 41.2$ and peaks at $R = 20$ at $t \approx 62$, before anything from the frozen core can arrive there; Sec. VII C bounds the freeze's imprint on the later record.

D. Fly-by burst

The $p = 0.45$ pass emits a single $(2, 2)$ arch of peak $|r \Psi_4| = 4.1 \times 10^{-2}$ with no ringdown—there is no remnant—propagating at $v/c = 1.00$ across every sphere pair; in retarded time all four spheres peak together at $t - R \simeq 34.5$. The record is complete to $t = 100$ with no NaN, but is gated at $t = 70$, 15.5 units after the $R = 20$ peak: the mouths' expansion (Sec. VII A) contaminates the innermost sphere, whose $|r \Psi_4|$ stops decaying at $t = 76.1$, where the contaminant has grown to equal the decaying burst.

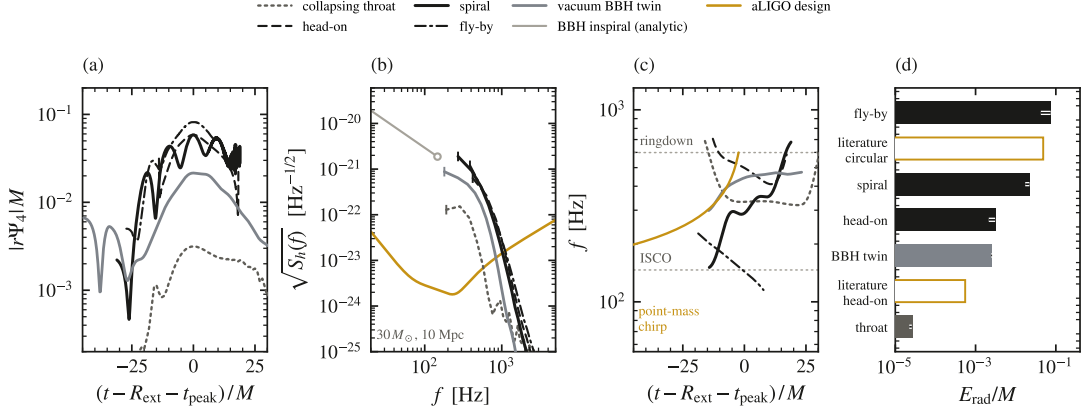


FIG. 8. Every source at one calibration: total mass $30 M_\odot$ at 10 Mpc, each record in units of its own total ADM mass. (a) Envelopes $|r\Psi_4|/M$ at the innermost sphere, aligned at each peak: fly-by loudest (8.2×10^{-2}) to lone throat (3.1×10^{-3}). (b) Strain amplitude spectral densities over the aLIGO design floor (gold), each from its own low-frequency corner (one cycle per record, ticked). The curves are falling power laws (slopes -3.4 to -5.8) with no resolved maximum, so any “peak” read here is the corner $1/T$. Faint line: the Newtonian inspiral of the same binary, $|\dot{h}| \propto f^{-7/6}$ to its ISCO at 147 Hz. (c) Instantaneous frequency against the point-mass chirp (gold), the ISCO (147 Hz) and the equal-mass Kerr ringdown (599 Hz): the vacuum twin climbs the chirp and peaks at $fM = 0.065$; no wormhole channel does (Sec. IX A). (d) Energy radiated in the dominant multipole; ticks give the spread over spheres (fly-by 43%, the widest). Open gold bars: published equal-mass vacuum values: head-on infall from rest at infinity, 5.5×10^{-4} [90]; quasi-circular inspiral from infinite separation and merger, 4.84×10^{-2} [84]; all multipoles. Energies are band integrals of $|\int r\Psi_4 dt'|^2$ over a common retarded window (the lone throat’s closing at $t = 58$, before its floor), doubled for $\pm m$; gravitational channel only. At 10^4 – $10^6 M_\odot$ and $z_e \simeq 20$ the same records are millihertz LISA sources (Sec. XB, Fig. 9).

E. Vacuum twin

The vacuum control—same d , p and masses, no scalar—behaves as a binary black hole must: it merges, climbs the point-mass chirp $299 \rightarrow 473$ Hz and turns toward the equal-mass Kerr mode, its Ψ_4 peak at the textbook $fM = 0.065$. At the spiral’s sphere the drainhole burst peaks $2.9\times$ higher than the twin’s merger and 43 units earlier (grey in Fig. 7(c)). The twin also calibrates the wavefront method: its flat-topped merger envelope defeats peak timing, which reads $0.57c$ on it against the $0.82c$ of its whole-waveform lag, so every speed here is a correlated lag.

F. Scalar channel and its sign

Ψ_4 is blind to a spin-0 wave. Two opposite scalar charges—every merging pair here (Sec. V)—carry a dipole, where a vacuum binary’s leading moment is the quadrupole. The scalar stream rides the same spheres as Ψ_4 on the spiral and fly-by, so the two channels compare sphere by sphere (Fig. 13 in Appendix B). The recorded kinematic flux F_{kin} (Sec. III C) approximates the scalar energy crossing S_R per coordinate time. That energy is the flux, through the world tube that S_R sweeps out, of the matter energy current $J^\mu \equiv -T^{(\phi)\mu}{}_\nu t^\nu$ of the time vector $t^\mu = \alpha n^\mu + \beta^\mu$, with $T_{\mu\nu}^{(\phi)}$ the canonical stress tensor (the negative of Eq. (1)). Since $\nabla_\mu J^\mu = -\frac{1}{2}T^{(\phi)\mu\nu}\mathcal{L}_t g_{\mu\nu}$, the current is conserved where t^μ is a

Killing vector and nearly so in the quasi-stationary exterior that the spheres sample, and Gauss’s theorem on the world tube gives its flux in 3+1 form [55, 56],

$$\begin{aligned} F_\phi^{\text{geo}} &= \oint_{S_R} \alpha s_i J^i \sqrt{\sigma} d^2x \\ &= - \oint_{S_R} \partial_t \phi (\alpha s^i \partial_j \phi + \beta_j s^j \Pi) \sqrt{\sigma} d^2x, \end{aligned} \quad (14)$$

positive for an outgoing canonical wave, with s^i the sphere’s outward unit normal, $\sqrt{\sigma}$ its area element, J^i the spatial components of the four-vector (not $\gamma^{ij} J_j$) and $\partial_t \phi = \alpha \Pi + \beta^i \partial_i \phi$. On a conformally flat sphere with uniform α and χ and no shift, $F_\phi^{\text{geo}} = \alpha^2 \chi^{-1/2} F_{\text{kin}}$; the shift adds an advective term $\propto (\beta^i \partial_i \phi)(s^j \partial_j \phi)$ and a term $\propto \beta_j s^j \Pi \partial_t \phi$ from its normal component. In the initial data the sphere at $R = 30$ has $\alpha \approx 0.93$ and $\chi \approx 0.87$, and the area mean of $\alpha^2 \chi^{-1/2}$ is ≈ 0.94 , so energies built from F_{kin} overstate $|F_\phi^{\text{geo}}|$ by about 6%; the shift terms are neglected.

The channel is dipole-dominated and comparable to the gravitational one. On the fly-by at $R = 30$, $\ell = 1$ carries the scalar power $\sum_{\ell m} |\dot{A}_{\ell m}|^2$ to one part in 10^7 , the $\ell = 1$ signal travels outward at c from $R = 14$, and the time-integrated flux to $t = 60$ is $|E_\phi| = 4.3 \times 10^{-2} M$ —2.3 times the gravitational energy that has crossed the sphere by then (Fig. 13(a)). The ratio depends on the cut and on how E_{GW} is integrated. With the running integral of Fig. 13(a) it falls to 1.1 by $t = 70$ and 0.8 by $t = 80$, but that integral keeps growing after the burst has passed, ac-

cumulating the drift of the time-integrated Ψ_4 ; the band-limited energy of Fig. 8(d) levels off instead (2.7, 1.5 and 1.4 at $t = 60, 70$ and 80). The wave-zone mode sum gives 1.7, and the inner sphere returns 3.0 for the same window. We therefore use only the channel’s multipole, direction, sign and order of magnitude downstream.

Gravity couples to minus the canonical stress tensor, Eq. (1), so the physical energy flux is $-F_{\text{kin}}$ and an outgoing scalar wave carries negative energy; what the data establish is that the $\ell = 1$ wave is outgoing (on the fly-by, and on the head-on after its horizon forms) and of the size above. We have not closed the ADM energy balance. If it holds with this term, a bound pair emitting it gains energy: the scalar channel would pump the orbit where gravitational radiation damps it, and radiation reaction would be *anti-damped*. This is the classical face of the standard objection to ghosts. Quantum mechanically a ghost’s vacuum decays into ghosts and positive-energy quanta together, at a rate kept finite only by cutting the effective theory off [74, 104]; classically these binaries make the same trade, emitting negative-energy scalar waves together with positive-energy gravitational waves, and the head-on remnant settles, within the accuracy of the scalar-flux estimate, near the pair’s Bondi mass, a little above $2M_{\text{ADM}}$ (Sec. VIB). The same bookkeeping appears at horizons, where swallowing the phantom shrinks the head-on remnant (Sec. VIB) and the single-throat remnant (Sec. IVE).

The dipole carries angular momentum with the same sign. On the fly-by, through both spheres and over its trusted record, the scalar’s physical angular-momentum flux is opposite to the gravitational waves’, which carry off angular momentum of the orbit’s sign: the channel that returns energy to the orbit returns angular momentum too. It also feeds the nonlinear memory, which the energy flux of any massless radiation sources [108–110]. The scalar’s flux enters with its negative sign, yet it adds to the gravitational (2, 0) memory rather than cancelling it: about the source’s axis, a dipole’s flux pattern projects onto that harmonic with the opposite sign to a quadrupole’s, and the negative energy turns it back. The fly-by and the head-on show the same sign on every sphere and window; the scalar’s share of the memory carries the near-zone uncertainty of E_ϕ above.

G. Horizon censorship of the scalar channel

The head-on, re-run at level 5 from $t = 0$ with the scalar stream, is the one encounter that forms a horizon early, and the horizon switches the channel off (Fig. 13 in Appendix B). After the common MOTS closes ($t = 21.5$ on this arm’s live scan), the mouths’ remaining hair leaves as a single $\ell = 1, m = 0$ pulse, cresting later at each sphere in causal order, and the envelope then decays: log-linear fits over $t = 30$ –95 give $\tau = 19/24/29$ at $R = 10/14/18$. The post-horizon energies keep the sign of Sec. VIIF, $E_\phi = -0.028/ -0.036/ -0.038 M$.

Only this window is integrated: before $t = 22$ the flux at $R = 10$ runs canonically ingoing, the two open mouths’ static hair superposing.

Without a horizon the channel does not shut. The fly-by’s dipole flux is still growing where its trusted record ends ($t - R = 50$). The spiral’s horizon, present from $t = 55.0$, is only five units old at its interior failure, against a shedding clock of ~ 20 , so it is too young to show the shut-off.

The lone throat shows the same contrast. Re-run at level 4 with the scalar stream, the $\varepsilon = +10^{-2}$ throat collapses on schedule—a permanent MOTS from $t = 11$, R settling $3.88 \rightarrow 2.54$ —and its scalar envelope, after the burst, falls by 30 – $100\times$ across the spheres by $t = 80$, with late e-fold times of 6 – 7 . The unkicked level-4 throat, which inflates with no trapped surface by scan or flow finder, instead grows quasi-exponentially in the $\ell = 0$ monopole (e-fold ≈ 5 , shortening across $t = 45$ – 75), identically in a box-doubled ($L = 128$) copy through $t \approx 80$, so the growth is the throat’s and not the boundary’s or the sponge’s; beyond $t \approx 80$ the constraint norms grow, and only $t \lesssim 80$ is counted. The contrast is confounded with the fate: the horizonless arms are also those whose mouths inflate toward the spheres. What is measured is that the scalar channel shuts off behind a horizon and grows while a throat inflates.

IX. PATTERN MATCHING AGAINST DETECTOR DATA

In 2.26 h of coincident Advanced LIGO O3b data [30], a matched-filter search [27, 29] with 127 templates—the waveforms of Sec. VIII rescaled as $h \propto M/D$, $f \propto 1/M$ —finds *no candidate at zero lag*, and none was expected. No static throat of astrophysical mass survives (Sec. IV G), so LIGO could catch only the burst of an encounter or collapse at formation, never a wormhole in residence; and a population formed at $z_e \gtrsim 20$ would convert there, far beyond the search’s reach (Sec. XB). At single-detector SNR 8 and optimal orientation the median horizon over each mass ladder is 4.8 Gpc (spiral), 2.5 (fly-by), 1.8 (head-on), 1.3 (vacuum twin) and 0.15 (collapsing throat); the first three are upper bounds, since their strain, from fixed-frequency integration [28], ends at 0.10–0.42 of peak. This short search demonstrates the pipeline; much stronger limits on short bursts follow from the LVK O3 all-sky burst search [85], whose rate-density limits against characteristic frequency apply directly to the energies of Sec. IX B.

The null result is not a blind spot. Cut to the signal’s duration, binary-black-hole templates (IMRPhenomD) match the bursts with fitting factors of 0.82–0.98, as they match the vacuum control (0.93–0.95), so burst morphology alone cannot tell the two apart. All 15 injections into O3b strain at optimal SNR 20 are recovered at 93–139% of optimal within 3 ms; only the lightest fly-by is demoted, by the χ^2 veto.

A. The frequency track

The frequency track separates a drainhole burst from a binary black hole (Fig. 8(c)). The vacuum twin peaks at $fM = 0.065$ and climbs the point-mass track before turning toward the equal-mass Kerr mode at $fM = 0.0885$ [82, 83]: a monotone rise, then a fixed ringdown frequency. No drainhole channel does this. The head-on’s frequency *falls*, and mass loss cannot explain it: a black hole shrinking as its remnant does rings faster, by $\times 1.08$ over the same window (with the area-based mass $R/2$), while the measurement is $\times 0.58$. The fly-by’s frequency falls as the pair separates; the spiral’s late record is carried by a frozen core and climbs without settling; the lone collapse rings at 0.55 of the equal-mass Kerr mode. A catalogue event [31] whose post-merger frequency holds within a few percent of a Kerr value is therefore inconsistent with the head-on channel at that mass.

B. Energy per event

The detector-independent deliverable is the energy per event in throat-mass units, in the dominant multipole (sphere spread in parentheses): fly-by $4.2 \times 10^{-2} M$ at the outermost sphere (7.4×10^{-2} at the innermost), spiral $2.2 \times 10^{-2} M$ (21%), head-on $3.1 \times 10^{-3} M$ (31%), collapsing throat $2.6 \times 10^{-5} M$ (14%), vacuum control $2.6 \times 10^{-3} M$ (5%). Since Ω_{GW} scales as event density times energy per event, these convert any bound on Ω_{GW} [25, 26] into a bound on the abundance of primordial drainhole encounters, provided the ghost scalar exists at that epoch. The scalar channel adds comparable energy of opposite sign (Sec. VIII F).

X. DISCUSSION

All four fates of Sec. I occur, each with a measured selection rule: the relative scalar sign separates scattering from merger (Sec. V), the seed’s sign separates collapse from inflation (Sec. IV), and the orbital momentum separates plunge from fly-by at 50–70% of circular (Sec. VII). For detection, three results combine: no inspiral exists (Sec. X A), the wormhole phase is a burst whose frequency falls or drifts where a Kerr remnant’s would hold (Sec. IX A), and the scalar counterpart decays once a horizon forms (Sec. VIII G). A surviving wormhole signature is a drifting-frequency burst with a scalar counterpart; a settled ringdown without one is already the descendant.

A. No wormhole inspiral

A drainhole binary cannot radiate an inspiral. Each throat’s unstable mode gives it a lifetime $t \approx \tau \ln(1/\varepsilon)$, with $\tau = 5.3$ –5.9 in isolation; in the binaries, once the

companion’s field is taken off the mouth reading, the fly-by’s throats grow faster ($\tau = 3.9$ over the fitted window) and the merging pair’s show no growth of their own before contact (Sec. VII C). An inspiral needs many orbits. Under the combined pull of Eq. (6), six times gravity at large separation and softened as $(d + \delta)^{-2}$ at the separations evolved (Sec. V B), a circular orbit at $d = 12$ has period $P \approx 75$ –100. On the isolated clock, the longest we measure, a throat carrying our superposed data’s effective seed, $\varepsilon_{\text{eff}} = 2 \times 10^{-4}$ – 6×10^{-4} , lasts about half a period. Even $\varepsilon = 10^{-15}$ buys only 1.8–2.7 periods, and each further factor of ten adds only $\tau \ln 10 \approx 12$ –14 units. The orbit’s demand grows with separation as $P \propto d^{3/2}$, while the throat’s endurance grows only as $\ln(1/\varepsilon)$. Constraint-solved initial data, lowering the seed to the level-3 truncation floor of 2×10^{-6} , would add $\tau \ln(\varepsilon_{\text{eff}}/2 \times 10^{-6}) \approx 25$ –34 units, under half a period.

Radiation reaction cannot close the gap either. The leading radiative moment is a scalar dipole carrying energy comparable to the gravitational channel’s, with negative sign (Sec. VIII F). If the ADM balance holds, it pumps the orbit rather than draining it, so there is no quasi-circular decay to compute, and the vacuum inspiral time does not apply.

The wormhole-specific signal is therefore a short transient: within a few tens of M each mouth collapses or inflates, and the plunge, burst and conversion of Secs. VI–VIII are all that this source class emits *as wormholes*. Three qualifications apply. The argument covers a minimally coupled ghost scalar without spin, the matter model of the González–Guzmán–Sarbach analysis, and holds only while the throat’s unstable mode exists. Rotation can remove that mode: Ellis wormholes set spinning exist up to an extremal-Kerr limit [57]; in five dimensions, with equal angular momenta, the static throat’s unstable mode disappears above a critical angular momentum [58]; in four, radial perturbations to second order in the rotation indicate the same [59, 60], and the non-axisymmetric modes computed at rapid rotation show no instability [61]. Other supports are untouched by the argument and include radially stable throats—thin shells for suitable equations of state [62] and dilatonic Einstein–Gauss–Bonnet wormholes, which need no exotic matter [63]—so a spinning or differently supported pair that outlives an inspiral is not excluded here. The same lifetimes rule out a static ghost-supported throat as one of the long-lived black-hole mimickers that ringdown and echo tests target, since those tests presuppose a wormhole that persists [105, 106]; only the rotating or differently supported throats above remain candidates. “No inspiral” means no *wormhole* inspiral: any long chirp belongs to the objects the throats become. And the seeds here are numerical, which the argument tolerates because its seed dependence is logarithmic and because the superposition defect’s excitation of the mode is measured: back-extrapolating the fly-by’s mouth growth gives this effective seed, 2×10^{-4} – 6×10^{-4} (with and without the companion’s field removed; Sec. VII C), just below the

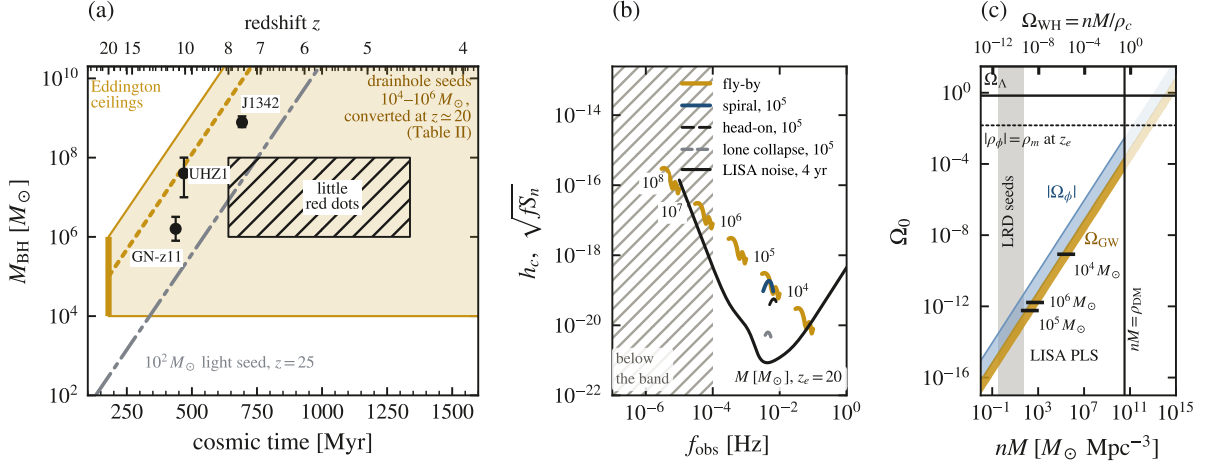


FIG. 9. The heavy-seed channel from measured numbers. (a) Seed growth: a drainhole population of 10^4 – $10^6 M_\odot$ converts at birth ($z \simeq 20$, seconds to minutes; Table IV), so its reachable region (gold) runs from the lightest seed idling to the heaviest seed’s Eddington ceiling (Salpeter e-fold 45 Myr). UHZ1 [46, 47], GN-z11 [50], the $z = 7.5$ quasar J1342+0928 [51] and the little-red-dot box [48, 49] lie inside it; the first three lie above the ceiling of a $10^2 M_\odot$ light seed started at $z = 25$ (grey), which misses UHZ1 by two decades. (b) One burst against LISA: the characteristic strain $h_c = 2f|\tilde{h}|$ of a fly-by (gold) at each mass decade from 10^4 to $10^8 M_\odot$ (exponents at the tracks), and of a spiral (blue), a head-on (dashed ink) and a lone collapse (grey) at $10^5 M_\odot$, all at $z_e = 20$, against $\sqrt{fS_n}$ of the analytic LISA noise [67] with the four-year galactic confusion (ink); inclination averaged, each track drawn over the band holding 90% of its $\int h_c^2 d \ln f$; the $10^7 M_\odot$ fly-by’s counted signal lies in its tail above the band floor, past the drawn core. The squared ratio of track to noise, integrated over $\ln f$, is the burst’s squared signal-to-noise ratio, so a short track can sit above the noise and still fall short of 8, as the lone collapse does. Heavier bursts are louder but redder: the tracks climb a decade per mass decade, the noise climbs faster below 0.1 mHz (hatched: below LISA’s band, not counted), and the $10^8 M_\odot$ burst lies wholly below it. At the abundances of (c) the population is ~ 0.05 –5 resolved bursts per year (Sec. XB). (c) Ω_{GW} of the conversions (gold, E/M from the head-on to the spiral) and the negative scalar deposit $|\Omega_{\phi,0}|$ (blue, $|E_\phi| = 0.8$ – $3.2 E_{\text{GW}}$ with E_{GW} from the spiral to the fly-by: every encounter, the case most favourable to Λ) today against the comoving mass density nM of converting wormholes, on which alone both depend (top axis: $\Omega_{\text{WH}} = nM/\rho_c$). Solid up to the dark-matter ceiling $nM = \rho_{\text{DM}}$ (vertical rule), faint beyond it: Ω_Λ is reached only at $\Omega_{\text{WH}} \simeq 60$ –800, above the cap $|\rho_\phi| = \rho_m$ at z_e that $H^2 > 0$ imposes (dashed). Ink ticks: the four-year power-law-integrated LISA sensitivity [68] at each mass’s conversion frequency $(fM)_{\text{peak}}/[M(1+z_e)]$. Grey: one seed per little red dot [48, 49] at 10^4 – $10^6 M_\odot$.

declared-kick decade over which the response is measured linear (Sec. IVC).

B. Heavy seeds and LISA bursts

The lifetimes of Table IV forbid any throat from surviving from the early universe, so everything below depends on the epoch z_e at which throats of mass M would form, which our evolutions do not fix. The mechanisms known to make macroscopic wormholes in the early universe—bubbles and domain walls [5, 6, 92, 93]—do so at horizon re-entry, $t \sim 1$ s for $10^5 M_\odot$ ($z \sim 10^9$ – 10^{10}), where these bursts would be redshifted out of every band. We know of no mechanism forming 10^4 – $10^6 M_\odot$ throats as late as $z_e \simeq 20$, so we present the consequences of such a population conditionally.

If such throats existed at $z_e \simeq 20$, collapse—alone or in a merger—would be a seeding mechanism. Black holes descended from wormholes have been proposed before. The wormholes that join the inflating interiors of super-

critical vacuum bubbles or domain walls to the exterior pinch off and leave black holes at their mouths [6, 92, 93], which for some parameter values can seed supermassive black holes [5, 92]; and some galactic nuclei have been proposed as former wormhole entrances [94]. Here the wormhole is traversable and ghost-supported, and its conversion is driven by its own measured instability or by a merger, at a remnant mass comparable to its own. A throat converts to a black hole in seconds at 10^4 – $10^5 M_\odot$ and in minutes at $10^6 M_\odot$ (Table IV), and the remnant keeps a mass of order its progenitors’: the head-on and spiral remnants both end at $R \simeq 4.16$ (irreducible mass 2.08) from pairs of ADM mass 2, and the lone collapse at $R \approx 2.3$ from a throat of mass 1 (Secs. IVE, VIB, VIIC). The scalar hair leaves on a $\sim 20 M$ clock after horizon formation (Sec. VIIG), so the remnants then accrete as ordinary black holes. From 10^4 – $10^5 M_\odot$ at $z \simeq 20$, Eddington-limited growth (Salpeter e-fold $\simeq 45$ Myr) reaches $10^9 M_\odot$ in 9–12 e-folds, 414–520 Myr, inside the ~ 600 Myr available by $z \simeq 7$: no super-Eddington phase is needed, which is what defines a heavy seed [52] (Fig. 9(a)). The channel would then seed JWST’s over-

massive black holes and little red dots [46–49] without an atomic-cooling halo, a critical Lyman–Werner flux, or any baryonic environment.

Halo independence would itself be a signature: the seeds would predate baryonic structure, so M_{BH}/M_* would start effectively infinite and relax downward, which would make UHZ1’s overmassive morphology the generic expectation rather than a tuning. The abundance required is small: one seed per present-day massive galaxy is $n \simeq 10^{-4}\text{--}10^{-2} \text{ Mpc}^{-3}$ comoving, a dark-matter fraction $f = nM/\rho_{\text{DM}} \simeq 3 \times 10^{-11}\text{--}3 \times 10^{-7}$ at $10^4\text{--}10^6 M_\odot$. The FIRAS μ -distortion bound [53], which excludes primordial black holes of $10^4\text{--}10^{13} M_\odot$ at any cosmologically relevant abundance, applies to black holes formed from Gaussian curvature perturbations; a wormhole-collapse channel is not subject to it, but only because its formation mechanism is left unspecified, as for the bubble and domain-wall channels [5, 6].

For $z_e \approx 20$ the encounters would radiate in LISA’s band (Fig. 9(b)). Each burst carries the energies of Sec. IX B at $f_{\text{obs}} = (fM)_{\text{peak}}/[M(1+z_e)]$; with the measured $(fM)_{\text{peak}} = 0.03\text{--}0.06$ the band is 0.3–60 mHz across $10^4\text{--}10^6 M_\odot$ at $z_e \simeq 20$: LISA’s band, not LIGO’s. One at a time the bursts would be loud. Rescaled with the strain of Sec. IX to $z_e = 20$ in a Planck cosmology [69] (detector-frame mass $(1+z_e)M$, luminosity distance ≈ 230 Gpc) and filtered against the analytic LISA noise [67], the fly-by and the spiral merger, the two loudest channels of Sec. VIII, reach optimal signal-to-noise ratios of 175–500 at $10^5\text{--}10^6 M_\odot$, and the head-on 60–79. All three stay above 8 from 4×10^4 to $6 \times 10^6 M_\odot$ with sky position and inclination averaged, the galactic confusion noise added and only each record’s resolved band counted, and carry the near-zone uncertainty of the energies (Sec. VIII); the strain pedestal of Sec. IX matters only at the light end. Heavier events are louder but redder, and LISA’s low-frequency noise rises faster than their strain (Fig. 9(b)): at $10^7\text{--}10^8 M_\odot$, where the bursts peak at 3–60 μHz , only the fly-by, whose spectrum extends furthest above its peak, still reaches 8, up to $2 \times 10^7 M_\odot$. The spiral and head-on bursts each announce a seed, since each ends in a black hole; the fly-by, the loudest, leaves two inflating mouths, announces none, and occurs at a rate not tied to the seeds’. A lone collapse, radiating only $2.6 \times 10^{-5} M$ (a background of $10^{-16}\text{--}10^{-14}$), stays below detection on the same count, at a signal-to-noise ratio of at most ≈ 6 (≈ 10 only at optimal orientation). What limits detection is occurrence: every seed converts once, so LISA records $4\pi D_c^2(z_e) c T n \simeq 1.9 \times 10^3 (n/\text{Mpc}^{-3})$ bursts in $T = 4$ yr, $\sim 0.05\text{--}5$ per year for $n = 10^{-4}\text{--}10^{-2} \text{ Mpc}^{-3}$, each $\sim 400 (M/10^5 M_\odot)$ s long. Averaged over time the conversions give $\Omega_{\text{GW}} \simeq n E_{\text{rad}}/[\rho_c(1+z_e)]$: with $E_{\text{rad}}/M = 3.1 \times 10^{-3}$ (head-on) to 2.2×10^{-2} (spiral) at $M = 10^5 M_\odot$, $n = 10^{-4}\text{--}10^{-2} \text{ Mpc}^{-3}$ gives $\Omega_{\text{GW}} \sim 10^{-14}\text{--}10^{-11}$ at ~ 6 mHz, above LISA’s four-year power-law-integrated sensitivity [68] ($\Omega \sim 10^{-13}\text{--}10^{-12}$ near millihertz [22, 24]) for $n \gtrsim (2\text{--}7) \times 10^{-4} \text{ Mpc}^{-3}$ at $10^6\text{--}10^5 M_\odot$ if every

conversion radiates like the spiral (Fig. 9(c)); fly-bys, which convert nothing, add up to $7.4 \times 10^{-2} M$ each at a rate the evolutions do not fix. At these rates, though, the signal is resolved bursts, never a stochastic background. The channel would therefore be detectable by LISA for merger conversions of $4 \times 10^4\text{--}6 \times 10^6 M_\odot$ with $n \gtrsim 5 \times 10^{-4} \text{ Mpc}^{-3}$, where at least one burst falls in four years, and not for lone collapses, for $10^4 M_\odot$ seeds or for $n \lesssim 10^{-4} \text{ Mpc}^{-3}$. The level and band scale as $1/(1+z_e)$, and $z_e \simeq 20$ is about the latest conversion epoch the seed arithmetic allows; a population converting at horizon re-entry would lie far below any band. Its spectrum is a pure burst population with no $f^{2/3}$ inspiral ramp (Sec. X A), unlike any direct-collapse scenario, whose seeds form hydrodynamically.

The negative-energy scalar radiation dilutes like radiation ($w = 1/3$) and has the wrong sign to act as dark energy; within the dark-matter budget its present density stays below 3×10^{-3} of critical.

Nothing in the astrophysical mass range survives (Sec. IV G), so every observed supermassive black hole is at most a descendant of this channel, not a surviving wormhole mouth [94, 95], and fits of a static Ellis–Bronnikov metric to the Sgr A* or M87* shadows are excluded within this model. The claim is conditional on z_e : collapse, alone or in a merger, would be a heavy-seed channel with one measured property—prompt conversion at comparable mass—two that would follow from an early birth—halo independence and freedom from the μ -distortion bound on curvature-perturbation black holes—and one testable signature: the encounter bursts, which LISA would resolve one by one. Whether it is *the* channel behind the JWST population is a question of abundance, and the abundance is cheap: one conversion per little red dot, $n \simeq (3\text{--}6) \times 10^{-5} \text{ Mpc}^{-3}$ (the number densities of Refs. [48, 49], summed over their bins), is a mass density $nM \simeq 0.3\text{--}60 M_\odot \text{ Mpc}^{-3}$ at $10^4\text{--}10^6 M_\odot$, a fraction $10^{-11}\text{--}2 \times 10^{-9}$ of the dark matter (Fig. 9(c)). LISA would then record 0.06–0.1 bursts in four years, so it tests the channel only if conversions outnumber the little red dots at least tenfold. The evolutions map a birth abundance $n(M)$ onto these observables; they do not fix it (Sec. X E).

C. The inflating half of the population

Seeding is half the census. The branch is chosen by the perturbation’s sign (Sec. IV C), so a sign-symmetric primordial seed distribution converts about half of a population of lone throats into black holes and sets the other half inflating; only large seeds skew the count, and toward collapse—at $\varepsilon = \pm 0.1$ both signs collapse (Sec. IV C). Pairs born close enough to collide (Sec. IV G) shift the count through the selection rules. Let a fraction $f_<$ of them be born below the capture threshold, at 50–70% of the momentum of a circular orbit (Sec. VII A). With the relative sign random, the like-signed half scatter, and

each of their throats then meets a lone throat’s fate if its own seed decides it; the opposite-signed half merge below the threshold and fly by above it, both mouths inflating. Per throat, a fraction $f_{<}/2$ then converts by merger, a quarter by lone collapse, and $3/4 - f_{<}/2$ inflates, so between a quarter and three quarters of a paired population ends in black holes, set by where its birth momenta fall about the threshold. An inflating throat creates, from regular data, an expanding anti-trapped region inside an asymptotically flat spacetime (Sec. IV F). Farhi and Guth showed that under the null energy condition such a region cannot be created classically without an initial singularity [71, 72]; the phantom violates that condition. Unlike the false-vacuum bubbles of Refs. [71, 72] and the domain-wall wormholes of Ref. [6], whose inflating interiors lie behind a black-hole horizon and pinch off, this region lies on our side of the throat and grows into the surrounding space, as the massless throat’s explosion does [13]. We therefore do not call it a baby universe. The record demonstrates the mechanism, not the magnitude: by $t = 218$ the neck accumulates 1.3 areal e-folds and the boundary 3.0, against the ~ 60 of cosmic inflation [73].

Whether such regions keep growing matters for any population: if half the throats inflate (Sec. IV C) and their anti-trapped regions grew without bound, they would eventually overlap. Our record cannot decide this. The outward motion of the $\theta_k = 0$ surface is measured in a slicing that freezes the neck, spans $218M$, and is evolved in an asymptotically flat rather than expanding background. The seed channel of Sec. XB therefore requires either that the growth stalls or that the inflating half is rare; an evolution in a slicing that does not freeze the neck is needed to decide. In the foam limit the mouths overlap by construction, and a predominantly inflating foam would be a candidate pre-inflationary expansion phase with a built-in exit, since growing perturbations widen the collapse basin; the equal-mass, two-throat evolutions here neither model such a foam nor fix its abundance.

D. Topological censorship

Topological censorship forbids an observer at infinity from probing the topology of a globally hyperbolic, asymptotically flat spacetime that satisfies the null energy condition: any topological structure collapses too quickly for light to traverse it [107]. A traversable wormhole evades the theorem only by violating that condition, and the two branches end the evasion in opposite ways. On the collapse branch, alone or in a merger, the throats end inside a MOTS while the phantom outside is swallowed or radiated away (Secs. IVE, VIB and VIIG). The wormholes are not removed—no classical evolution changes the number of asymptotic ends [16], and both pits stay distinct inside the spiral’s remnant (Sec. VII C)—but they end behind a horizon, hidden from

outside as the theorem would have them. On the inflation branch no trapped surface forms and the phantom is not swallowed: the throat stays open while its anti-trapped region grows (Sec. IV F), and nothing restores the condition. The sign of the seed, or the encounter, decides which way a throat goes.

E. Scope

The results hold for one matter model and one family of data: a single minimally coupled ghost scalar, equal masses, one throat compactness ($m/a = 1/2$) and no spin, on constraint-solved, matched initial data whose momentum is the exact boost (Secs. IID and IIE). Horizons here are MOTSs on the evolved slices, which for matter violating the null energy condition need not lie inside event horizons (Sec. IIID). The ghost field is treated classically [74] (Sec. I). The evolved binary has three asymptotic ends, not a handle’s two mouths. Time-step convergence is not tested, the growth rates rest on two resolutions, waveforms are extracted in the near zone without extrapolation, and the ADM energy balance including the scalar channel is not closed. The cosmology of Sec. XB takes the birth abundance $n(M)$ and the conversion epoch z_e as inputs—Fig. 9(c) maps any choice onto its signals—and requires the ghost scalar to exist at that epoch.

XI. REPRODUCIBILITY AND COST

Every evolution in Table III is one line in a registry shipped with the results pack—its single knob against a named reference, its outcome, its caveats—and every figure is regenerated by a script in the pack from the archived diagnostic streams. The code, the analysis and the results pack are in the `GRTEclyn` [41] fork at <https://github.com/Nikchik-coder/GRTEclyn> (branch `feature/merger`): the drainhole initial data, the seeds of Sec. IIF, the interior freeze and the in-code diagnostics in the evolution code; the flow finder, the orientation-corrected surface scan, the figure scripts and the claims ledger that recomputes every number quoted here in the analysis layer; and the parameter files and thinned diagnostic streams of every run in the pack. Each run’s manifest names the binary it ran, and a table in the pack maps every binary to its source commit; the campaign binary from 24 September 2026 on is commit 7166787a, which it carries stamped. The evolutions are bit-reproducible on the production hardware: two builds differing only in a passive diagnostic give byte-identical physics streams over 2384 output rows.

Every evolution ran on one NVIDIA H100 80GB device, with `GRTEclyn` built CUDA-native against AM-ReX 26.02. At $\Delta t = 0.02 \Delta x$ a production arm ($L = 64$, 128^3 base, three refinement levels) advances 6–18 code units per hour, and a $t = 100$ single-throat or head-on

evolution takes 5–7 h. Depth is the expense: level-5 arms run at 2–4 units per hour on the full box (the octant $L = 512$ arm, an eighth of the cells, at 28–74, 11.9 h in all)—29.3 h for the head-on’s three-leg $t = 0 \rightarrow 100$ chain, 46.3 h for the $p = 0.25$ fly-by on the doubled box, which peaks at 58 of the 80 GB, and 26.9 h for the $d = 6$ merger chain. The 204 physics evolutions total 995 GPU-hours: 646 recomputed from the pack’s logs (each run’s logged speed and time span, a restarted leg from its checkpoint) and 349 for the archived superposed binaries, measured the same way before archiving; the shakedown adds 9 h.

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Appendix A: Code Health and Constraint Evolution

This appendix collects the checks that the evolutions are numerically sound, so that the main-text figures show physics only. Figure 10 gives the constraint record of every run drawn in Figs. 2, 3, 3, 5 and 6, each at the highest refinement level it was run at. Figure 11 shows the spiral’s interior failure under refinement, gauge and start time (Sec. VII D); its bottom half checks the linearity of the collapsing throat’s wave in its quadrupole seed (Fig. 11; Sec. VIII A).

Appendix B: Supplementary Measurements

Three measurements support the main text without being needed to follow it: the momentum scan that separates plunge from scattering (Fig. 12; Sec. VII A), the scalar channel’s energy, multipoles and validity window on the Ψ_4 spheres (Fig. 13; Sec. VIII F), and the horizon’s shut-off of that channel (Fig. 13; Sec. VIII G).

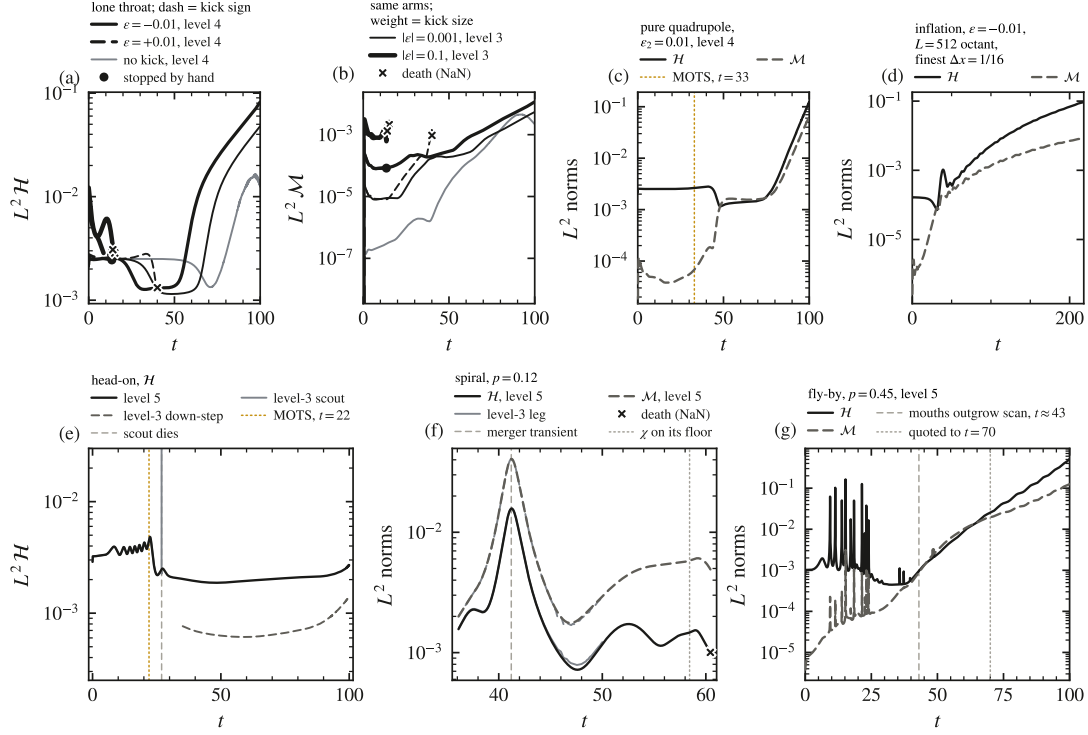


FIG. 10. Constraint evolution of every run the main-text figures draw, each at the highest refinement level it was run at (named in each panel's key): L^2 norms of the Hamiltonian ($\mathcal{H}$, ink) and momentum ($\mathcal{M}$, grey dashed) constraints. The norms are box averages on the base grid ($\Delta x = 0.5$; 2 on the octant of (d)) and compare only within one box (Sec. III C). (a),(b) The lone throat of Fig. 2: level 4 for the unkicked throat and $\varepsilon = \pm 10^{-2}$ ($+10^{-2}$ stopped by hand at $t = 13.5$, dot), level 3 for $\pm 10^{-3}$ and ± 0.1 , the only level these were run at (dash = sign, weight = amplitude, grey = no kick, crosses = deaths). The base grid does not resolve the seed's shell: at $t = 0$ the norms carry the exact data's discretisation floor, to which a kick adds its own defect in quadrature, $0.090 |\varepsilon|$ ($2.1\text{--}2.35 \times 10^{-3}$ for $|\varepsilon| \leq 10^{-2}$), although on the shell it reaches $0.93 \times 16\pi |\rho|$ at 10^{-2} (Sec. II F). $\mathcal{M}$ vanishes at $t = 0$ and is fed by that defect within the first steps, in proportion to ε (2.3×10^{-4} at 10^{-2} , 2.3×10^{-5} at 10^{-3} , at $t = 0.5$, against 2×10^{-7} unkicked). At level 3 the $|\varepsilon| \leq 10^{-2}$ arms then follow the unkicked norms to 5.4% until the fates part—a bound on the box average, not on the violation near the throat (Sec. IV E); $\varepsilon = \pm 0.1$ starts 3–6 $\times$ high and dies at $t = 14.1$ ($+0.1$) and 15.2 (-0.1) in a one-step overflow at the compactified origin. (c) The pure-quadrupole collapse of Fig. 3 (level 4; gold, its MOTS from $t = 33$). The rise from $t \gtrsim 75$ is a grid mode, neither this collapse nor a reflection off the box face: it starts on the refinement boundaries, every single-throat arm that runs long enough shows it, collapsing or inflating, and the doubled box does not delay it: the norms leave their floor from $t \approx 77$ ($L = 128$) and 82 ($L = 64$). (d) The inflating throat of Fig. 3 (finest $\Delta x = 1/16$ on the $L = 512$ octant) to $t = 218$. (e) The head-on of Fig. 5, $\mathcal{H}$ only: the level-5 arm stays below its value at formation (gold, $t = 22$) to $t = 100$, while the level-3 scout's norm leaves the frame at its failure ($t = 26.91$, grey dashed); dashed, the level-3 down-step. (f) The production spiral of Fig. 6: the level-5 leg to its NaN at $t = 60.45$ (cross), and the level-3 leg over their common window, which it reproduces to a median 3.5%/0.7%; rules, the merger transient ($t = 41.2$) and χ on its floor ($t = 58.4$). At the NaN both norms sit below their transient peaks: the failure is not a constraint blow-up. (g) The fly-by of Sec. VII A ($p = 0.45$, level 5, $L = 128$): $\mathcal{H}$ stays flat between thin early needles (isolated spikes, each gone within about a tenth of a time unit) until the mouths outgrow the scan window ($t \approx 43$, first rule); both norms then grow by orders of magnitude to the end of the record, and nothing is quoted from the run after $t = 70$ (second rule).

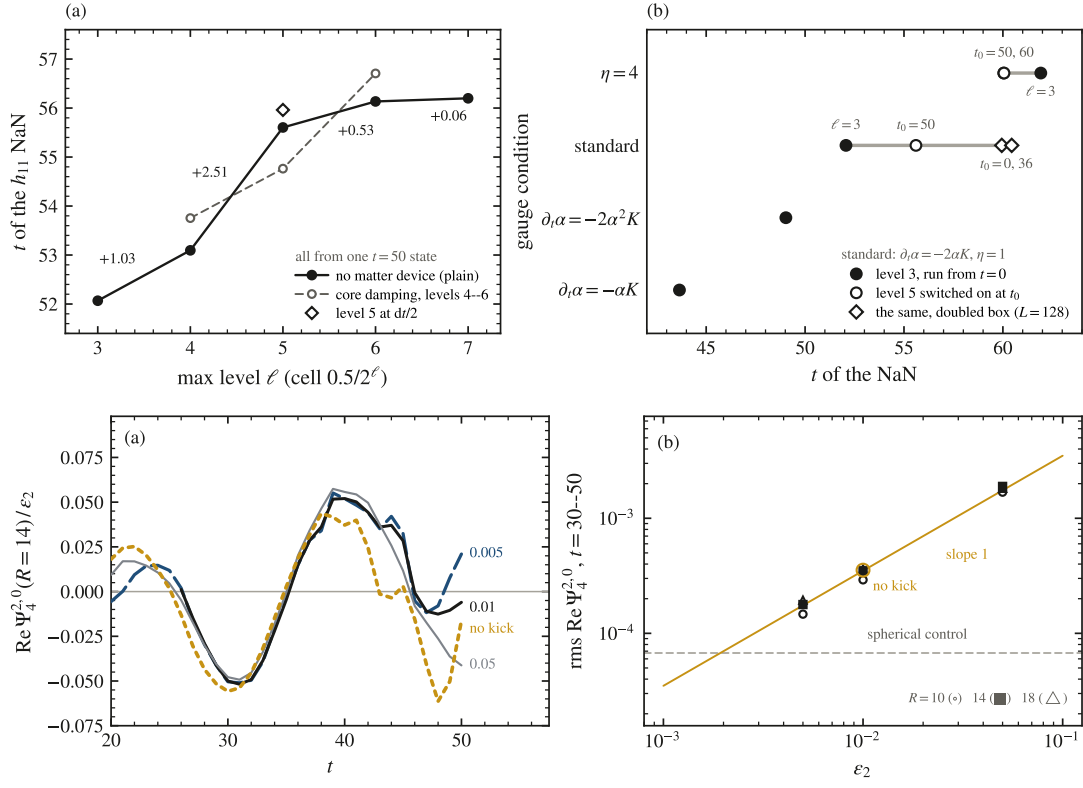


FIG. 11. *Top (refinement ladder)*: Death time of the $p = 0.12$ spiral under refinement, gauge and start time; every death is an h_{11} NaN except in the $\partial_t \alpha = -\alpha K$ arm, which fails in K . (a) Refinement from one $t = 50$ state of the $L = 64$ run, maximum level ℓ (finest cell $0.5/2^\ell$). Filled circles: no matter device, gain per doubling marked; open circles: core matter damping (levels 4–6 share one configuration); open diamond: level 5 at half the time step. (b) The same failure under other gauges and start times, one row per gauge. The lapse obeys $\partial_t \alpha = -c\alpha^p(K - 2\Theta)$ and the shift is damped at rate η ; the standard gauge is $c = 2$, $p = 1$, $\eta = 1$, and each other row changes one of the three (and carries the core matter damping, worth under a unit; panel a). Filled circles: level 3 from $t = 0$; open circles: level 5 from t_0 on $L = 64$; open diamonds: level 5 on $L = 128$. Level 3: $t = 43.6, 49.0, 52.1, 61.9$. Level 5: standard gauge 59.94 and 60.45 on $L = 128$ ($t_0 = 0, 36$) and 55.60 on $L = 64$ ($t_0 = 50$); $\eta = 4$, 60.04 and 60.05 ($t_0 = 50, 60$). *Bottom (seed linearity)*: Linearity of the collapsing throat's wave in its quadrupole seed. (a) $\text{Re } \Psi_4^{2,0}$ at $R = 14$, each arm divided by its seed and named at its end: $\varepsilon_2 = 0.005, 0.01, 0.05$ and the kick-free 0.01 arm collapse onto one curve through the burst; later the smallest arm carries its floor amplified by $1/\varepsilon_2$ and the kick-free arm runs ahead. (b) rms amplitude over $t = 30-50$ against ε_2 at three spheres, with the slope-one line through the 0.01 arm drawn across the decade $10^{-3}-10^{-1}$ as a reference. Open circle: the pure quadrupole, within 2% of the kicked arm. Dashed rule: the floor of the level-4 spherical control (the same kick, no quadrupole) at $R = 14$, which the line crosses at $\varepsilon_2 \approx 2 \times 10^{-3}$. The outcome: the wave is the quadrupole's—linear in ε_2 and unchanged, to 2% in amplitude, when the radial kick is removed. The kick alone radiates only the floor; it moves the wave's phase, not its size.

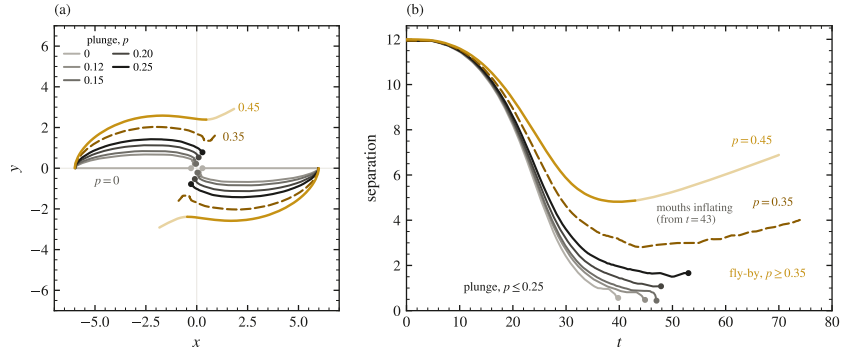


FIG. 12. Merger or fly-by on boosted, solved data: the $d = 12$ momentum scan and the $d = 6$ merger. (a) Both throats in the orbital plane; (b) their separation. The tracks are the coordinate positions of the χ pits, followed through each run by continuity and smoothed over two time units; a track ends (dot) where its pits stop being trackable (χ floored, or a superluminal hop between two minima) or at the run's trust window. Gold: the scatterers, $p = 0.25$ (solid, to $t = 63.3$) and 0.45 (dashed, to $t = 67.6$). Ink and greys: the plunges $p = 0.60$ and 0.90 (two legs each, to $t = 55.5$ and 45.3), the $d = 6$, $p = 0.10$ merger, and the $p = 0.12$ arm, whose mouths inflate and whose pits merge near $t = 38$ without a horizon ever forming.

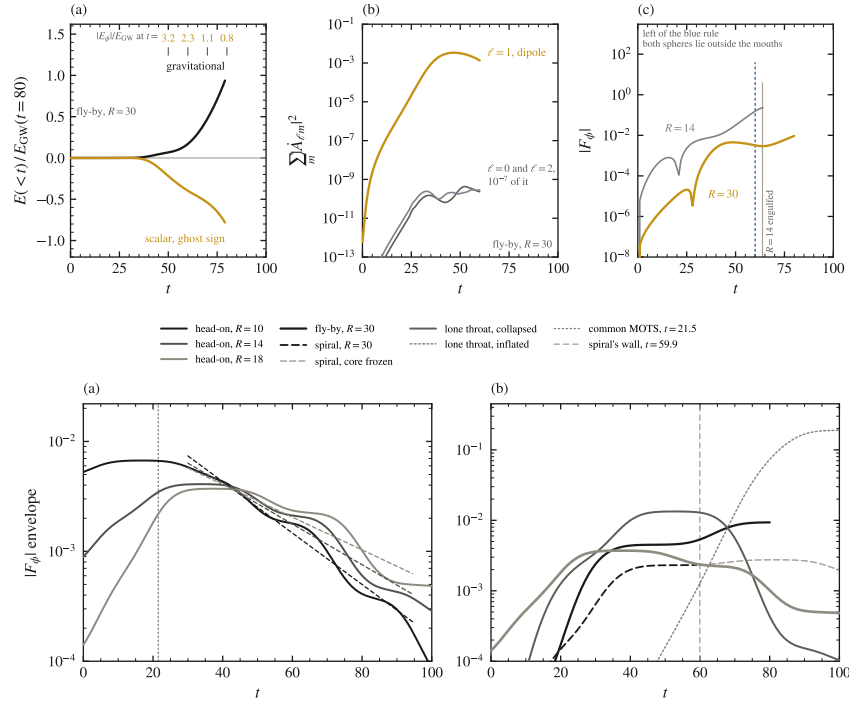


FIG. 13. *Top (channel)*: The scalar channel on the Ψ_4 spheres. (a) Fly-by energy through $R = 30$ in units of the gravitational energy crossed by $t = 80$: gravitational (ink) and scalar with its physical sign (gold, Sec. VIII F). The ratio $|E_\phi|/E_{\text{GW}}$ printed at each cut falls from 3.2 to 0.8 as the gravitational burst arrives and this running integral drifts (Sec. VIII F); $t = 80$ is where the fly-by's trust window closes on this sphere ($t - R = 50$). (b) Multipole content, fly-by: $\ell = 1$ exceeds $\ell = 0$ and $\ell = 2$ by 10^7 . (c) Validity window: $|F_\phi|$ at both spheres, each drawn to $t - R = 50$ (the fly-by's trust window); the grey rule marks where the mouths' areal-radius reading (an upper bound) passes the inner sphere, which within the window it never does for the outer one; left of the blue rule both spheres lie outside the mouths. The $t = 60$ window at $R = 14$ gives 3.0 in place of 2.3. *Bottom (censorship)*: Envelope of the scalar flux $|F_\phi|$, horizoned against horizonless arms: a running maximum over 25 units (one dipole period), smoothed in log amplitude, since the raw flux changes sign every ~ 20 units. (a) The head-on (level 5 from $t = 0$) at $R = 10/14/18$. The common MOTS closes at $t = 21.5$ (dotted); the remaining hair leaves as one $\ell = 1$ pulse, cresting later at each sphere, then decays (dashed: fits over $t = 30-95$, $\tau = 19/24/29$). The rise before the rule is the open mouths' static hair, canonically ingoing; only the post-horizon window is integrated. (b) Each encounter on its outermost sphere (head-on and lone throat $R = 18$, fly-by and spiral $R = 30$): the fly-by, drawn to $t - R = 50$ (its trust window), is still rising there; the spiral, with a horizon from $t = 55.0$ and too young to shed on a ~ 20 clock, is still climbing at its interior failure ($t = 59.9$, dashed rule), beyond which the frozen-core arm is drawn dashed and carries no claim. Mid-grey: the lone throat twice—the $\varepsilon = +10^{-2}$ level-4 collapse (solid), behind a MOTS from $t = 11$, and the unkicked level-4 throat (dotted), which inflates, horizonless, and grows in the monopole (counted to $t \approx 80$).

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- [1] J. A. Wheeler, “Geons,” *Phys. Rev.* **97**, 511 (1955).
 - [2] J. A. Wheeler, “On the nature of quantum geometrodynamics,” *Ann. Phys. (N.Y.)* **2**, 604 (1957).
 - [3] C. W. Misner and J. A. Wheeler, “Classical physics as geometry,” *Ann. Phys. (N.Y.)* **2**, 525 (1957).
 - [4] T. A. Roman, “Inflating Lorentzian wormholes,” *Phys. Rev. D* **47**, 1370 (1993).
 - [5] J. Garriga, A. Vilenkin, and J. Zhang, “Black holes and the multiverse,” *J. Cosmol. Astropart. Phys.* **02**, 064 (2016).
 - [6] H. Deng, J. Garriga, and A. Vilenkin, “Primordial black hole and wormhole formation by domain walls,” *J. Cosmol. Astropart. Phys.* **04**, 050 (2017).
 - [7] A. A. Kirillov and E. P. Savelova, “Dark matter from a gas of wormholes,” *Phys. Lett. B* **660**, 93 (2008).
 - [8] A. A. Kirillov and E. P. Savelova, “Density perturbations in a gas of wormholes,” *Mon. Not. R. Astron. Soc.* **412**, 1710 (2011).
 - [9] H. G. Ellis, “Ether flow through a drainhole: A particle model in general relativity,” *J. Math. Phys.* **14**, 104 (1973).
 - [10] K. A. Bronnikov, “Scalar-tensor theory and scalar charge,” *Acta Phys. Polon. B* **4**, 251 (1973).
 - [11] M. S. Morris and K. S. Thorne, “Wormholes in spacetime and their use for interstellar travel: A tool for teaching general relativity,” *Am. J. Phys.* **56**, 395 (1988).
 - [12] M. Visser, *Lorentzian Wormholes: From Einstein to Hawking* (AIP Press, Woodbury, NY, 1995).
 - [13] H. Shinkai and S. A. Hayward, “Fate of the first traversible wormhole: Black-hole collapse or inflationary expansion,” *Phys. Rev. D* **66**, 044005 (2002).
 - [14] J. A. González, F. S. Guzmán, and O. Sarbach, “Instability of wormholes supported by a ghost scalar field. II. Nonlinear evolution,” *Class. Quantum Grav.* **26**, 015011 (2009).
 - [15] J. A. González, F. S. Guzmán, and O. Sarbach, “Instability of wormholes supported by a ghost scalar field. I. Linear stability analysis,” *Class. Quantum Grav.* **26**, 015010 (2009).
 - [16] R. Geroch, “Topology in general relativity,” *J. Math. Phys.* **8**, 782 (1967).
 - [17] F. J. Tipler, “Singularities and causality violation,” *Ann. Phys. (N.Y.)* **108**, 1 (1977).
 - [18] S. W. Hawking, “Wormholes in spacetime,” *Phys. Rev. D* **37**, 904 (1988).
 - [19] Ya. B. Zel’dovich and I. D. Novikov, “The hypothesis of cores retarded during expansion and the hot cosmological model,” *Sov. Astron.* **10**, 602 (1967).
 - [20] B. J. Carr and S. W. Hawking, “Black holes in the early Universe,” *Mon. Not. R. Astron. Soc.* **168**, 399 (1974).
 - [21] B. Carr, K. Kohri, Y. Sendouda, and J. Yokoyama, “Constraints on primordial black holes,” *Rep. Prog. Phys.* **84**, 116902 (2021).
 - [22] C. Caprini and D. G. Figueroa, “Cosmological backgrounds of gravitational waves,” *Class. Quantum Grav.* **35**, 163001 (2018).
 - [23] G. Agazie *et al.* (NANOGrav Collaboration), “The NANOGrav 15 yr data set: Evidence for a gravitational-wave background,” *Astrophys. J. Lett.* **951**, L8 (2023).
 - [24] P. Amaro-Seoane *et al.*, “Laser Interferometer Space Antenna,” arXiv:1702.00786 [astro-ph.IM] (2017).
 - [25] R. Abbott *et al.* (LIGO Scientific, Virgo and KAGRA Collaborations), “Upper limits on the isotropic gravitational-wave background from Advanced LIGO and Advanced Virgo’s third observing run,” *Phys. Rev. D* **104**, 022004 (2021).
 - [26] L. Pagano, L. Salvati, and A. Melchiorri, “New constraints on primordial gravitational waves from Planck 2015,” *Phys. Lett. B* **760**, 823 (2016).
 - [27] B. Allen, W. G. Anderson, P. R. Brady, D. A. Brown, and J. D. E. Creighton, “FINDCHIRP: An algorithm for detection of gravitational waves from inspiraling compact binaries,” *Phys. Rev. D* **85**, 122006 (2012).
 - [28] C. Reisswig and D. Pollney, “Notes on the integration of numerical relativity waveforms,” *Class. Quantum Grav.* **28**, 195015 (2011).
 - [29] S. A. Usman *et al.*, “The PyCBC search for gravitational waves from compact binary coalescence,” *Class. Quantum Grav.* **33**, 215004 (2016).
 - [30] R. Abbott *et al.* (LIGO Scientific, Virgo and KAGRA Collaborations), “Open data from the third observing run of LIGO, Virgo, KAGRA and GEO,” *Astrophys. J. Suppl.* **267**, 29 (2023).
 - [31] R. Abbott *et al.* (LIGO Scientific, Virgo and KAGRA Collaborations), “GWTC-3: Compact binary coalescences observed by LIGO and Virgo during the second part of the third observing run,” *Phys. Rev. X* **13**, 041039 (2023).
 - [32] C. W. Misner and D. H. Sharp, “Relativistic equations for adiabatic, spherically symmetric gravitational collapse,” *Phys. Rev.* **136**, B571 (1964).
 - [33] D. Hochberg and M. Visser, “Dynamic wormholes, antitrapped surfaces, and energy conditions,” *Phys. Rev. D* **58**, 044021 (1998).
 - [34] S. W. Hawking, “Gravitational radiation from colliding black holes,” *Phys. Rev. Lett.* **26**, 1344 (1971).
 - [35] J. M. Bowen and J. W. York, “Time-asymmetric initial data for black holes and black-hole collisions,” *Phys. Rev. D* **21**, 2047 (1980).
 - [36] T. Helfer, U. Sperhake, R. Croft, M. Radia, B.-X. Ge, and E. A. Lim, “Malaise and remedy of binary boson-star initial data,” *Class. Quantum Grav.* **39**, 074001 (2022).
 - [37] D. Alic, C. Bona-Casas, C. Bona, L. Rezzolla, and C. Palenzuela, “Conformal and covariant formulation of the Z4 system with constraint-violating solutions,” *Phys. Rev. D* **85**, 064040 (2012).
 - [38] C. Bona, J. Massó, E. Seidel, and J. Stela, “New formalism for numerical relativity,” *Phys. Rev. Lett.* **75**, 600 (1995).
 - [39] M. Campanelli, C. O. Lousto, P. Marronetti, and Y. Zlochower, “Accurate evolutions of orbiting black-hole binaries without excision,” *Phys. Rev. Lett.* **96**, 111101 (2006).
 - [40] J. G. Baker, J. Centrella, D.-I. Choi, M. Koppitz, and J. van Meter, “Gravitational-wave extraction from an inspiraling configuration of merging black holes,” *Phys. Rev. Lett.* **96**, 111102 (2006).
 - [41] GRTL Collaboration, *GRTeclyn*, <https://github.com/GRTLCollaboration/GRTeclyn>.

- [42] W. Zhang *et al.*, “AMReX: a framework for block-structured adaptive mesh refinement,” *J. Open Source Softw.* **4**, 1370 (2019).
- [43] T. Andrade *et al.*, “GRChombo: an adaptable numerical relativity code for fundamental physics,” *J. Open Source Softw.* **6**, 3703 (2021).
- [44] N. M. Shirokov, “Wormhole dynamics: non-linear collapse and gravitational-wave emission,” arXiv:2604.00071 [gr-qc] (2026).
- [45] N. M. Shirokov, “The Bondi dipole in full numerical relativity: a self-accelerating positive-negative mass binary,” arXiv:2608.24577 [gr-qc] (2026).
- [46] Á. Bogdán *et al.*, “Evidence for heavy-seed origin of early supermassive black holes from a $z \approx 10$ X-ray quasar,” *Nat. Astron.* **8**, 126 (2024).
- [47] P. Natarajan, F. Pacucci, A. Ricarte, Á. Bogdán, A. D. Goulding, and N. Cappelluti, “First detection of an overmassive black hole galaxy: UHZ1—evidence for heavy black hole seeds from direct collapse?,” *Astrophys. J. Lett.* **960**, L1 (2024).
- [48] J. E. Greene *et al.*, “UNCOVER spectroscopy confirms the surprising ubiquity of active galactic nuclei in red sources at $z > 5$,” *Astrophys. J.* **964**, 39 (2024).
- [49] J. Matthee *et al.*, “Little red dots: an abundant population of faint active galactic nuclei at $z \sim 5$ revealed by the EIGER and FRESCO JWST surveys,” *Astrophys. J.* **963**, 129 (2024).
- [50] R. Maiolino *et al.*, “A small and vigorous black hole in the early Universe,” *Nature* **627**, 59 (2024).
- [51] E. Bañados *et al.*, “An 800-million-solar-mass black hole in a significantly neutral Universe at a redshift of 7.5,” *Nature* **553**, 473 (2018).
- [52] K. Inayoshi, E. Visbal, and Z. Haiman, “The assembly of the first massive black holes,” *Annu. Rev. Astron. Astrophys.* **58**, 27 (2020).
- [53] T. Nakama, B. Carr, and J. Silk, “Limits on primordial black holes from μ distortions in cosmic microwave background,” *Phys. Rev. D* **97**, 043525 (2018).
- [54] M. Alcubierre, “Hyperbolic slicings of spacetime: singularity avoidance and gauge shocks,” *Class. Quantum Grav.* **20**, 607 (2003).
- [55] K. Clough, “Continuity equations for general matter: applications in numerical relativity,” *Class. Quantum Grav.* **38**, 167001 (2021).
- [56] E.ourgoulhon, *3+1 Formalism in General Relativity: Bases of Numerical Relativity*, Lecture Notes in Physics Vol. 846 (Springer, Berlin, 2012).
- [57] B. Kleihaus and J. Kunz, “Rotating Ellis wormholes in four dimensions,” *Phys. Rev. D* **90**, 121503(R) (2014).
- [58] V. Dzhunushaliev, V. Folomeev, B. Kleihaus, J. Kunz, and E. Radu, “Rotating wormholes in five dimensions,” *Phys. Rev. D* **88**, 124028 (2013).
- [59] B. Azad, J. L. Blázquez-Salcedo, F. S. Khoo, and J. Kunz, “Are slowly rotating Ellis-Bronnikov wormholes stable?,” *Phys. Lett. B* **848**, 138349 (2024).
- [60] B. Azad, J. L. Blázquez-Salcedo, F. S. Khoo, and J. Kunz, “Radial perturbations of Ellis-Bronnikov wormholes in slow rotation up to second order,” *Phys. Rev. D* **109**, 124051 (2024).
- [61] F. S. Khoo, B. Azad, J. L. Blázquez-Salcedo, L. M. González-Romero, B. Kleihaus, J. Kunz, and F. Navarro-Lérída, “Quasinormal modes of rapidly rotating Ellis-Bronnikov wormholes,” *Phys. Rev. D* **109**, 084013 (2024).
- [62] E. Poisson and M. Visser, “Thin-shell wormholes: Linearization stability,” *Phys. Rev. D* **52**, 7318 (1995).
- [63] P. Kanti, B. Kleihaus, and J. Kunz, “Wormholes in dilatonic Einstein-Gauss-Bonnet theory,” *Phys. Rev. Lett.* **107**, 271101 (2011).
- [64] K. A. Bronnikov, J. C. Fabris, and A. Zhidenko, “On the stability of scalar-vacuum space-times,” *Eur. Phys. J. C* **71**, 1791 (2011).
- [65] S. A. Hayward, “Gravitational energy in spherical symmetry,” *Phys. Rev. D* **53**, 1938 (1996).
- [66] E. Babichev, V. Dokuchaev, and Yu. Eroshenko, “Black hole mass decreasing due to phantom energy accretion,” *Phys. Rev. Lett.* **93**, 021102 (2004).
- [67] T. Robson, N. J. Cornish, and C. Liu, “The construction and use of LISA sensitivity curves,” *Class. Quantum Grav.* **36**, 105011 (2019).
- [68] E. Thrane and J. D. Romano, “Sensitivity curves for searches for gravitational-wave backgrounds,” *Phys. Rev. D* **88**, 124032 (2013).
- [69] N. Aghanim *et al.* (Planck Collaboration), “Planck 2018 results. VI. Cosmological parameters,” *Astron. Astrophys.* **641**, A6 (2020).
- [70] A. Ashtekar and B. Krishnan, “Isolated and dynamical horizons and their applications,” *Living Rev. Relativ.* **7**, 10 (2004).
- [71] E. Farhi and A. H. Guth, “An obstacle to creating a universe in the laboratory,” *Phys. Lett. B* **183**, 149 (1987).
- [72] S. K. Blau, E. I. Guendelman, and A. H. Guth, “Dynamics of false-vacuum bubbles,” *Phys. Rev. D* **35**, 1747 (1987).
- [73] A. R. Liddle and S. M. Leach, “How long before the end of inflation were observable perturbations produced?,” *Phys. Rev. D* **68**, 103503 (2003).
- [74] J. M. Cline, S. Jeon, and G. D. Moore, “The phantom menaced: Constraints on low-energy effective ghosts,” *Phys. Rev. D* **70**, 043543 (2004).
- [75] S. A. Hayward, “General laws of black-hole dynamics,” *Phys. Rev. D* **49**, 6467 (1994).
- [76] S. A. Hayward, “Dynamic wormholes,” *Int. J. Mod. Phys. D* **8**, 373 (1999).
- [77] C. Gundlach, “Pseudospectral apparent horizon finders: An efficient new algorithm,” *Phys. Rev. D* **57**, 863 (1998).
- [78] M. Alcubierre *et al.*, “Gauge conditions for long-term numerical black hole evolutions without excision,” *Phys. Rev. D* **67**, 084023 (2003).
- [79] G. Huisken and T. Ilmanen, “The inverse mean curvature flow and the Riemannian Penrose inequality,” *J. Differ. Geom.* **59**, 353 (2001).
- [80] M. Mars, “Present status of the Penrose inequality,” *Class. Quantum Grav.* **26**, 193001 (2009).
- [81] T. Damour, F. Guercilena, I. Hinder, S. Hopper, A. Nagar, and L. Rezzolla, “Strong-field scattering of two black holes: Numerics versus analytics,” *Phys. Rev. D* **89**, 081503(R) (2014).
- [82] E. Berti, V. Cardoso, and C. M. Will, “Gravitational-wave spectroscopy of massive black holes with the space interferometer LISA,” *Phys. Rev. D* **73**, 064030 (2006).
- [83] E. Berti, V. Cardoso, and A. O. Starinets, “Quasinormal modes of black holes and black branes,” *Class. Quantum Grav.* **26**, 163001 (2009).
- [84] M. A. Scheel, M. Boyle, T. Chu, L. E. Kidder, K. D. Matthews, and H. P. Pfeiffer, “High-accuracy

- waveforms for binary black hole inspiral, merger, and ringdown,” *Phys. Rev. D* **79**, 024003 (2009).
- [85] R. Abbott *et al.* (LIGO Scientific, Virgo and KAGRA Collaborations), “All-sky search for short gravitational-wave bursts in the third Advanced LIGO and Advanced Virgo run,” *Phys. Rev. D* **104**, 122004 (2021).
 - [86] C. Gao, X. Chen, V. Faraoni, and Y.-G. Shen, “Does the mass of a black hole decrease due to the accretion of phantom energy?,” *Phys. Rev. D* **78**, 024008 (2008).
 - [87] A. Doroshkevich, J. Hansen, I. Novikov, and A. Shatskiy, “Passage of radiation through wormholes,” *Int. J. Mod. Phys. D* **18**, 1665 (2009).
 - [88] F. Cremona, F. Pirodda, and L. Pizzocchero, “On the linear instability of the Ellis–Bronnikov–Morris–Thorne wormhole,” *Gen. Relativ. Gravit.* **51**, 19 (2019).
 - [89] S. A. Hayward, “Wormhole dynamics in spherical symmetry,” *Phys. Rev. D* **79**, 124001 (2009).
 - [90] H. Witek, M. Zilhão, L. Gualtieri, V. Cardoso, C. Herdeiro, A. Nerozzi, and U. Sperhake, “Numerical relativity for D dimensional space-times: Head-on collisions of black holes and gravitational wave extraction,” *Phys. Rev. D* **82**, 104014 (2010).
 - [91] I. D. Novikov and D. I. Novikov, “Collapse of a wormhole and its transformation into black holes,” *J. Exp. Theor. Phys.* **129**, 495 (2019).
 - [92] H. Deng and A. Vilenkin, “Primordial black hole formation by vacuum bubbles,” *J. Cosmol. Astropart. Phys.* **12**, 044 (2017).
 - [93] Y. Gouttenoire and E. Vitagliano, “Primordial black holes and wormholes from domain wall networks,” *Phys. Rev. D* **109**, 123507 (2024).
 - [94] N. S. Kardashev, I. D. Novikov, and A. A. Shatskiy, “Astrophysics of wormholes,” *Int. J. Mod. Phys. D* **16**, 909 (2007).
 - [95] C. Bambi, “Can the supermassive objects at the centers of galaxies be traversable wormholes? The first test of strong gravity for mm/sub-mm very long baseline interferometry facilities,” *Phys. Rev. D* **87**, 107501 (2013).
 - [96] C. Gundlach and J. M. Martín-García, “Critical phenomena in gravitational collapse,” *Living Rev. Relativ.* **10**, 5 (2007).
 - [97] M. W. Choptuik, T. Chmaj, and P. Bizoń, “Critical behavior in gravitational collapse of a Yang-Mills field,” *Phys. Rev. Lett.* **77**, 424 (1996).
 - [98] R. Penrose, “Gravitational collapse: The role of general relativity,” *Riv. Nuovo Cimento* **1**, 252 (1969) [*Gen. Relativ. Gravit.* **34**, 1141 (2002)].
 - [99] R. M. Wald, “Gravitational collapse and cosmic censorship,” in *Black Holes, Gravitational Radiation and the Universe*, edited by B. R. Iyer and B. Bhawal (Springer, Dordrecht, 1999), p. 69.
 - [100] C. Palenzuela, P. Pani, M. Bezares, V. Cardoso, L. Lehner, and S. Liebling, “Gravitational wave signatures of highly compact boson star binaries,” *Phys. Rev. D* **96**, 104058 (2017).
 - [101] M. Bezares, C. Palenzuela, and C. Bona, “Final fate of compact boson star mergers,” *Phys. Rev. D* **95**, 124005 (2017).
 - [102] M. Bezares, M. Bošković, S. Liebling, C. Palenzuela, P. Pani, and E. Barausse, “Gravitational waves and kicks from the merger of unequal mass, highly compact boson stars,” *Phys. Rev. D* **105**, 064067 (2022).
 - [103] C. Palenzuela, I. Olabarrieta, L. Lehner, and S. L. Liebling, “Head-on collisions of boson stars,” *Phys. Rev. D* **75**, 064005 (2007).
 - [104] S. M. Carroll, M. Hoffman, and M. Trodden, “Can the dark energy equation-of-state parameter w be less than -1 ?,” *Phys. Rev. D* **68**, 023509 (2003).
 - [105] V. Cardoso, E. Franzin, and P. Pani, “Is the gravitational-wave ringdown a probe of the event horizon?,” *Phys. Rev. Lett.* **116**, 171101 (2016); **117**, 089902(E) (2016).
 - [106] P. Bueno, P. A. Cano, F. Goelen, T. Hertog, and B. Vercnocke, “Echoes of Kerr-like wormholes,” *Phys. Rev. D* **97**, 024040 (2018).
 - [107] J. L. Friedman, K. Schleich, and D. M. Witt, “Topological censorship,” *Phys. Rev. Lett.* **71**, 1486 (1993); **75**, 1872(E) (1995).
 - [108] D. Christodoulou, “Nonlinear nature of gravitation and gravitational-wave experiments,” *Phys. Rev. Lett.* **67**, 1486 (1991).
 - [109] K. S. Thorne, “Gravitational-wave bursts with memory: The Christodoulou effect,” *Phys. Rev. D* **45**, 520 (1992).
 - [110] M. Favata, “Post-Newtonian corrections to the gravitational-wave memory for quasi-circular, inspiralling compact binaries,” *Phys. Rev. D* **80**, 024002 (2009).